

Indian Mineral Import Policy Options

An Economywide Analysis



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INDIAN MINERAL IMPORT POLICY OPTIONS: AN ECONOMYWIDE ANALYSIS

- Dr. Badri Narayanan Gopalakrishnan
Vishnu Das Gupta
Kannan Kumar

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ENDORSEMENT

The mineral and mining sector plays a vital role in supporting India's industrial growth and economic development. As the country continues to expand its infrastructure and manufacturing base, the strategic management of mineral resources—particularly in the context of imports—has become increasingly important.

Indian Mineral Import Policy Options: An Economywide Analysis, authored by **Dr. Badri Narayanan Gopalakrishnan, Vishnu Das Gupta, and Kannan Kumar**, provides a timely and insightful examination of this critical area. The authors bring together empirical analysis and economy-wide modelling to offer a clear understanding of India's mineral import dynamics and their broader economic implications. The work highlights the balance India must maintain between domestic resource utilization and external dependence.

What makes this volume particularly valuable is its policy orientation. Evaluating the impact of alternative scenarios on growth, trade, and employment, it offers a useful framework for informed decision-making. The study will be of relevance to policymakers, researchers, and industry stakeholders alike.

I commend the authors for their rigorous and thoughtful contribution to this important field. I am confident that this work will enrich ongoing discussions on resource policy and economic strategy in India.



Prof. Dr. Tamma Koti Reddy

Vice Chancellor (In-charge)

ICFAI Foundation for Higher Education (IFHE), Hyderabad

FOREWORD

It is a privilege and an honor to write the foreword for this insightful and analytically rich work titled *“Indian Mineral Import Policy Options: An Economywide Analysis.”* This publication arrives at a critical juncture when India’s mineral sector is undergoing significant transformation, driven by evolving policy frameworks, increasing industrial demand, and the need for sustainable resource management.

The mineral and mining sector forms the backbone of India’s industrial and economic development. It provides essential raw materials that fuel core industries such as steel, aluminum, cement, power, and infrastructure. As highlighted in this study, the sector not only contributes meaningfully to the nation’s Gross Value Added (GVA) but also plays a vital role in employment generation and regional development. India, being endowed with substantial mineral resources, has traditionally maintained a degree of self-sufficiency in several key minerals. However, the growing demand for specific minerals and metals, coupled with quality constraints and technological requirements, has led to a persistent reliance on imports for certain critical inputs.

In this context, the present work assumes considerable importance. The authors have undertaken a comprehensive and methodologically sound analysis of India’s mineral import dynamics, covering a wide spectrum of minerals including iron ore, coal, copper, bauxite, precious metals, and rare minerals. The study systematically examines trends in imports, global trade patterns, tariff structures, and domestic production capabilities, thereby offering a holistic understanding of the sector.

One of the most notable strengths of this publication is its rigorous analytical framework. The use of economy wide modeling and simulation techniques to assess the impact of changes in imports and domestic production provides valuable insights into the broader macroeconomic implications. Such an approach enables policymakers and stakeholders to evaluate not only sector-specific outcomes but also their cascading effects on employment, wages, trade balances, and overall economic growth. This is particularly relevant in today’s interconnected economic environment, where policy decisions in one sector often have far-reaching consequences across the economy.

From a financial and strategic perspective, the issues addressed in this book are of paramount significance. The import of minerals has direct implications on foreign exchange outflows, cost structures of downstream industries, and the competitiveness of Indian manufacturing in global markets. At the same time, domestic production is influenced by factors such as resource availability, regulatory frameworks, technological capabilities, and investment climate. Striking an optimal balance between imports and domestic production is therefore crucial for ensuring both economic efficiency and resource security.

The study also sheds light on the comparative tariff structures of major mineral-importing countries, highlighting India’s position in the global landscape. Such comparative analysis is essential for informed policy formulation, as it provides a benchmark against international practices and helps identify areas where policy adjustments may be warranted. In an era of increasing global competition and trade integration, aligning domestic policies with global best practices becomes imperative.

Another important dimension of this work is its emphasis on policy evolution and regulatory reforms. The mining sector in India has witnessed significant changes in recent years, including amendments to the Mines and Minerals (Development and Regulation) Act, the introduction of transparent auction mechanisms, and the implementation of the National Mineral Policy 2019. These reforms aim to enhance transparency, attract investment, and promote sustainable mining practices. The book effectively contextualizes these developments and examines their implications for the sector's future trajectory.

As a finance professional associated with a leading public sector enterprise in the mining and metals domain, I find the insights presented in this publication both relevant and thought-provoking. The intersection of mineral economics, trade policy, and financial sustainability is an area of growing importance, particularly in the context of India's aspiration to become a global manufacturing hub. Efficient resource utilization, prudent financial planning, and evidence-based policymaking are essential for achieving this objective.

Furthermore, the discussion on import dependence for minerals where India possesses significant reserves raises important questions about supply chain efficiencies, value addition, and industrial strategy. Addressing these issues requires a coordinated approach involving government, industry, and research institutions. This book provides a strong analytical foundation for such discussions and can serve as a valuable reference for decision-makers.

The clarity of presentation, supported by extensive data, tables, and graphical analysis, enhances the readability and practical utility of the work. The authors have succeeded in striking a balance between academic rigor and policy relevance, making the book accessible to a diverse audience, including policymakers, industry practitioners, academicians, and students.

In conclusion, I commend the authors for their meticulous research, analytical depth, and commitment to addressing a subject of national importance. This publication is a timely contribution to the discourse on India's mineral policy and will undoubtedly aid in shaping informed and effective policy decisions. It is my firm belief that this work will stimulate constructive dialogue and encourage further research in this critical area.

I extend my best wishes to the authors and hope that this book reaches a wide audience and contributes meaningfully to strengthening India's mineral and economic policy framework.



Abhimanyu Padhi

CA Abhimanyu Padhi
Deputy General Manager (Finance)
NALCO, Bhubaneswar

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उप महाप्रबंधक (वित्त) / DGM (Finance)
नेशनल एल्युमिनियम कंपनी लिमिटेड
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PREFACE

Minerals and metals constitute the backbone of industrial development and economic transformation. They are indispensable inputs across key sectors such as infrastructure, manufacturing, energy, and advanced technologies. For a rapidly developing economy like India, ensuring a secure, resilient, and efficient supply of mineral resources is of strategic importance.

India's critical mineral supply chain, however, faces several structural challenges. These include limited domestic availability of certain resources, high-risk and capital-intensive exploration processes, long gestation periods for mining projects, and the geographical concentration of processing capabilities and reserves. Additionally, aggressive fiscal and policy support extended by major economies to secure critical minerals further intensifies global competition. In this context, the formulation of well-informed and forward-looking mineral import policies becomes an essential component of India's broader economic and industrial strategy.

Against this backdrop, *Indian Mineral Import Policy Options: An Economywide Analysis* offers a timely and insightful examination of India's mineral trade dynamics and their wider economic implications. While India possesses significant reserves of minerals such as iron ore, bauxite, chromite, and coal—and ranks among the leading global producers—it continues to depend on imports for several critical minerals and metals that are vital for industrial growth and technological advancement. This dual reality highlights the need for a carefully calibrated balance between domestic production and strategic imports.

A key strength of this volume lies in its comprehensive analysis of India's mineral import patterns over time. The authors examine long-term trends across a diverse set of minerals, including iron ore, limestone, manganese, aluminium, copper, gold, silver, rock phosphate, nickel, and coal. By integrating trade data with insights from global commodity markets, the study provides a nuanced understanding of the economic forces shaping India's mineral imports and the evolving structure of domestic demand.

An important contribution of the book is its application of economy-wide modelling techniques to assess the impact of changes in mineral import policies. Through simulation exercises focusing on critical minerals such as copper, tungsten, and nickel, the authors evaluate how variations in imports and domestic production can influence output, employment, prices, and overall economic activity. Such rigorous analysis offers valuable inputs for policymakers seeking to design balanced, evidence-based resource strategies.

The book also situates India's tariff structure within a global comparative framework, highlighting differences in import duties across major mineral-importing countries. This perspective raises important considerations regarding competitiveness, trade policy design, and the role of tariffs in shaping the mineral value chain.

Furthermore, the discussion of recent reforms in India's mining sector—including developments under the Mines and Minerals (Development and Regulation) framework and the National Mineral Policy—provides important institutional context. These reforms aim to enhance transparency, attract investment, and promote sustainable mining practices, thereby supporting the long-term development of the sector.

Overall, this volume represents a well-researched and policy-relevant contribution to the field of mineral economics and trade policy. By combining empirical analysis with broader economic insights, the authors offer a robust framework for understanding the challenges and opportunities associated with India's mineral import strategies.

I commend the authors for their rigorous and thoughtful work, and I am confident that this book will stimulate informed debate and further research on the role of mineral resources in India's economic development—particularly at a time when global supply chains are undergoing significant transformation.



Prof. Dheeraj Kumar

Deputy Director &

Head, Centre of Excellence on National Critical Mineral Mission

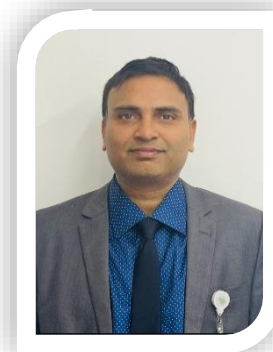
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Dr. Badri Narayanan Gopalakrishnan is a globally renowned economist specializing in international trade, economic policy, and global value chains. He is affiliated with leading academic institutions including the University of Washington, Oregon State University, and Boston College, and serves on the board of the Global Trade Analysis Project (GTAP) at Purdue University.

He has worked with premier institutions such as the World Bank, International Monetary Fund, and Asian Development Bank. As a Fellow at NITI Aayog, he has significantly contributed to shaping India's trade and economic policies. With over 15 books and 250+ research papers, his work has had a global policy impact.



Vishnu Das Gupta

Vishnu Das Gupta is an independent economist who has worked in the areas of macroeconomics, public policy, international trade policy, economic modeling, and development economics.

His expertise lies in macroeconomics, trade and industrial policy. He brings a strong understanding of policy formulation and implementation, bridging the gap between governance and industry needs. His insights in trade policy and industrial competitiveness make him a valuable voice in India's evolving economic landscape.



Kannan Kumar

Kannan Kumar is an independent economist and advocate specializing in trade policy and industrial policy. His work spans economic research, policy analysis, and advisory engagements across key areas of India's economic development.

He has contributed to significant national policy initiatives and international trade negotiations, bringing together rigorous empirical analysis and a strong understanding of the legal and regulatory frameworks that shape economic outcomes.



ACKNOWLEDGEMENTS

The completion of this book has been a deeply enriching and collaborative journey, made possible through the invaluable support, intellectual contributions, and encouragement of many individuals and institutions. We express our sincere gratitude to all those who contributed directly and indirectly to shaping this work.

At the outset, we extend our heartfelt appreciation to the dedicated researchers whose assistance significantly strengthened this book: **Ritu Baid, Aman Vedanth Agarwal, Palak Sharma, Supriya, Ishita Tripathi, Anish Uppal, and Ayush Raj**. Their contributions included conducting literature reviews, compiling datasets, refining drafts, validating arguments, and bringing diverse perspectives with academic rigor and commitment to excellence.

We especially acknowledge **Ritu Baid** for her meticulous attention to detail and ability to synthesize complex information; **Aman Vedanth Agarwal** for his analytical depth and policy-oriented insights; **Palak Sharma** for her structured approach and consistency; **Supriya** for her thoughtful inputs and dedication; **Ishita Tripathi** for her enthusiasm and research acumen; **Anish Uppal** for his critical thinking and constructive feedback; and **Ayush Raj** for his proactive engagement and timely research support.

We are equally grateful to the distinguished policy leaders, domain experts, and seminar participants from India and abroad whose valuable feedback on earlier drafts enhanced the quality and relevance of this manuscript. Their discussions and critiques helped us challenge assumptions and refine our analysis.

In particular, we sincerely thank **Dr. Rajesh Chadha** and **Ganesh Sivamani** from the Centre for Social and Economic Progress (CSEP). Their insightful comments, grounded in academic rigor and practical policy experience, were instrumental in strengthening the analytical framework of this book.

We also acknowledge the broader academic and policy community whose research, policy papers, and scholarly discussions provided an important foundation for this study. Our appreciation extends to all stakeholders and seminar participants whose questions and suggestions offered fresh perspectives and ensured the relevance of this work.

On a personal note, we thank our families and close associates for their unwavering support, patience, and encouragement throughout this demanding journey.

Finally, any errors or omissions that remain are solely our responsibility. This book stands as a testament to the power of collaboration, collective knowledge, and shared purpose. We hope it contributes meaningfully to ongoing discussions and serves as a valuable resource for researchers, practitioners, and policymakers alike.

— **Dr. Badri Narayanan Gopalakrishnan, Vishnu Das Gupta, Kannan Kumar**

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CHAPTER | 1

INTRODUCTION

India's mineral economy is at a very important point right now because of the growing demand for minerals in industry and the country's reliance on trade with other countries. This chapter lays the groundwork by looking at the sector's structural strengths and weaknesses and emphasizing how important minerals are to India's economic growth.

IN THIS CHAPTER...

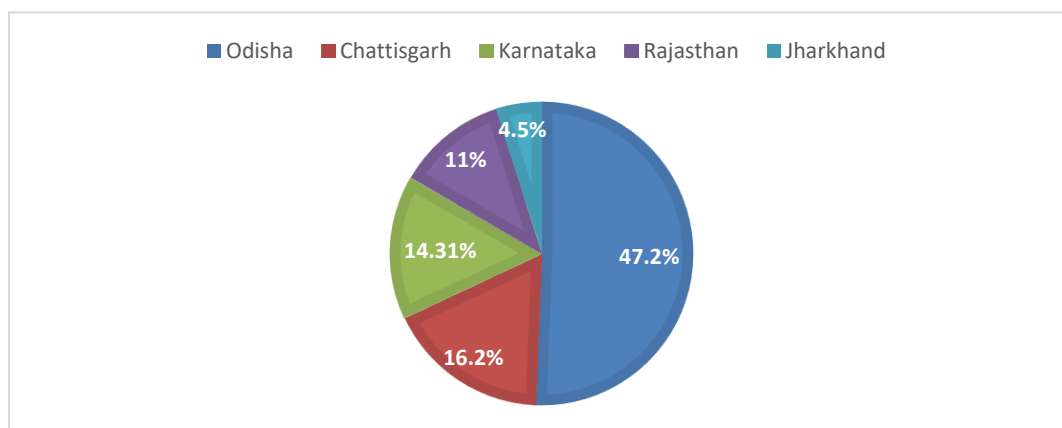
- Overview of India's mineral sector
- Important minerals and their role in the economy
- The relationship between demand and supply
- Rules and policies for governance

Mining plays a significant role in India's economy, contributing approximately 1.63% to the country's Gross Value Added (GVA) during 2020–21, according to provisional estimates. This sector supplies essential raw materials that support a wide range of manufacturing and infrastructure industries across the nation. India possesses abundant reserves of both metallic and non-metallic minerals. The country has around 1,531 active mines and produces a total of 95 different minerals. These include 4 fuel minerals, 10 metallic minerals, 23 non-metallic minerals, 3 atomic minerals, and 55 minor minerals such as building materials and others. Geological surveys have identified about 571,000 square kilometres of land—out of India's total area of 3.1 million square kilometres—as having Obvious Geological Potential (OGP). These regions are considered highly prospective for the presence of mineral deposits.

In 2021–22, mineral production was recorded across 21 States and Union Territories in India. However, a significant concentration of output was observed, with nearly 88.7% of the total value of mineral production excluding fuel, atomic, and minor minerals coming from just four states. Among these, Odisha held the top position, contributing approximately 47.2% of the country's total mineral output by value. It was followed by Chhattisgarh with a 16.2% share. Karnataka

accounted for 14.31%, while Rajasthan contributed around 11%, and Jharkhand made up about 4.5% of the overall mineral production value.

Figure 1: Share of estimated value mineral production (percentage)



Source: Ministry of Mines- Annual Report 2021-22

India has remained fully or largely self-reliant in several minerals that serve as essential raw materials for key industries such as iron and steel, aluminium, cement, refractories, ceramics based on China clay, and glass manufacturing. The country is either self-sufficient or nearly so in minerals like bauxite, chromite, iron ore, and limestone. However, India faces shortages in certain minerals, including kyanite, magnesite, manganese ore, rock phosphate, sillimanite, tungsten, and nickel. To meet industrial requirements, these minerals are imported either to supplement domestic supplies or to produce specialized, high-quality mineral-based products.

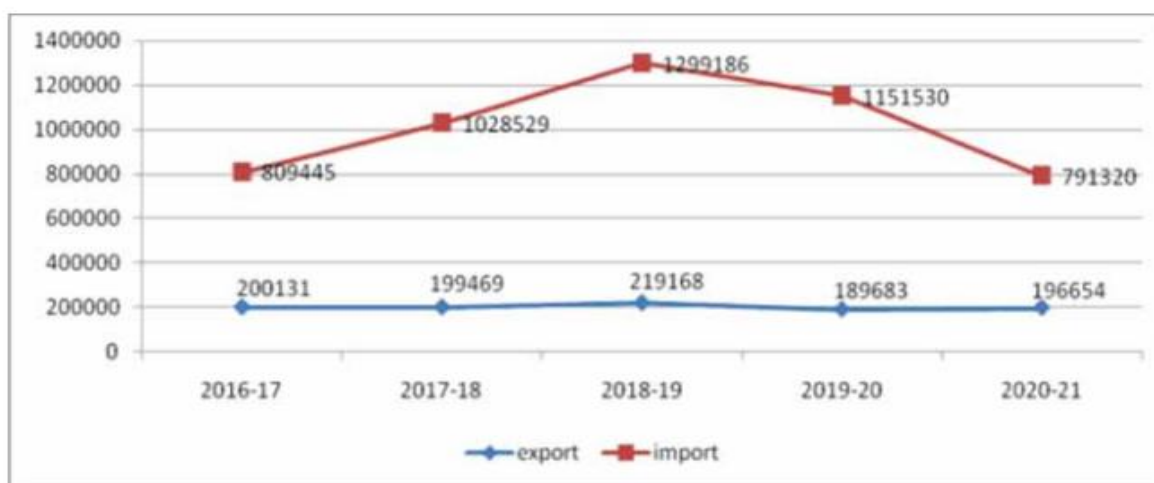
Table 1: Demand, Supply and Self Sufficiency of Minerals and Metals

S.No.	Commodity	Demand/Domestic Consumption(tonnes)	Supply/Domestic supply(tonnes)	Order of self-sufficiency (%)
Minerals				
1	Bauxite	22189	23688	100
2	Chromite	1920	3971	100
3	Iron ore	159940	206446	100
4	Kyanite	5.1	4.89	96
5	Limestone	350878	379045	100
6	Magnesite	195	147	75
7	Manganese ore*	5548	2820	51
8	Rock phosphate (including apatite)	8802	1285	15
9	Sillimanite	56	69	100

Metals				
1	Aluminium (primary)	3676	3696	100
2	Copper (refined)	1159	454	39
3	Lead (primary)	381	198	52
4	Zinc	778	696	89

Source: Ministry of Mines- Annual Report 2021-22

Figure 2: Trends in Value of Mineral Exports & Imports (INR Crore)



Source: Ministry of Mines-Annual Report 2021-22

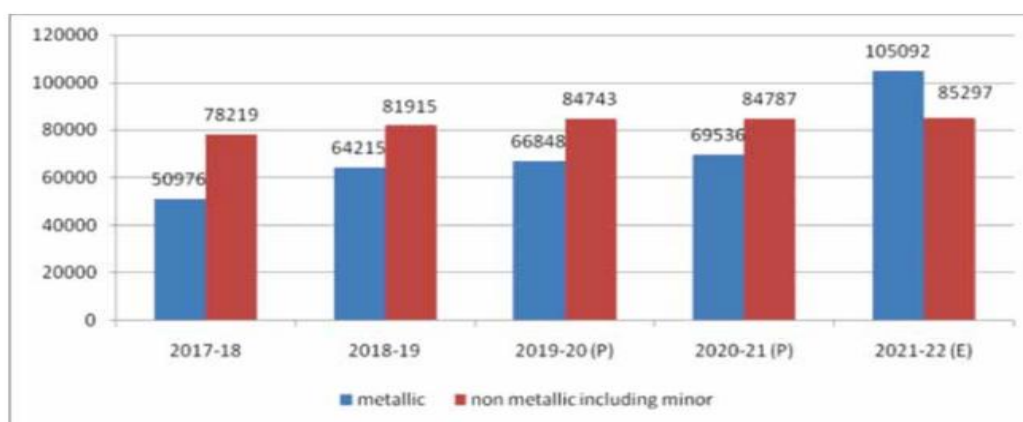
Table 2: Contribution and Rank of India in World Production of Principal Minerals & Metals, 2019

Sector	Unit of Commodity	Production(quantity)		Contribution (Percentage)	India's rank in World order ^s
		World	India*		
Metallic Minerals					
Bauxite	'000 tonnes	347100	21824	6.29	5 th
Chromite	'000 tonnes	38600	3929	10.18	3 rd
Iron ore	million tonnes	3040	246	8.09	4 th
Manganese ore	'000 tonnes	56600	2904	5.13	7 th
Industrial Minerals**					
Magnesite	'000 tonnes	29700	98	0.33	17 th
Apatite & rock phosphate	'000 tonnes	226000	1400	0.62	16 th
Metals					
Aluminium (Primary)	'000 tonnes	62900	3635	5.78	3 rd
Copper (refined)	'000 tonnes	24100	408	1.69	13 th
Steel (crude/liquid)	million tonnes	1814	109.13	5.89	2 nd
Lead (refined) ^{e##}	'000 tonnes	12500	599 [#]	4.79	4 th
Zinc (slab)	'000 tonnes	13500	516	3.82	4 th

Source: Ministry of Mines-Annual Report 2021-22

In 2021–22, the total value of mineral production in India excluding atomic, fuel, and minor minerals was estimated at ₹1,90,389 crore, reflecting an increase of about 23.37% compared to the previous year. Out of this, metallic minerals accounted for ₹1,05,092 crore, contributing 55.2% of the total value, while non-metallic minerals (including minor minerals) were valued at ₹85,297 crore, making up the remaining 44.8%.

Figure 3: Value of Minerals Production (by groups) (excluding atomic and fuel minerals)



Source: Ministry of Mines-Annual Report 2021-22

India is among the leading global producers of key minerals such as chromite, iron ore, coal, and bauxite. The mining sector in the country is strictly regulated, and in the past five years, the legal framework has undergone significant reforms, resulting in a more transparent and efficient system.

Constitutional provisions related to the Sector

India's mineral and mining sector functions within a federal system, where regulatory authority is shared between the central government and individual State governments. This division of powers is defined by the Union List, State List, and Concurrent List outlined in the Seventh Schedule of the Constitution of India.

The administration of India's mining sector is a shared responsibility between the central and State governments, as provided by the Constitution. Under Entry 54 of the Union List, the central government is empowered to regulate mines and mineral development when such regulation is declared by Parliament to be in the public interest. Meanwhile, the authority of State governments under Entry 23 of the State List to oversee mines and mineral development remains subject to the central government's powers. In line with its authority under the Union List, the central government enacted the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act), which serves as the primary law governing the mineral sector in India, excluding petroleum and natural gas.

Recent Developments

The Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act), was amended in 2015, along with the introduction of new rules, to eliminate discretionary practices and enhance transparency in the allocation of mineral concessions. Under the amended provisions, such concessions are granted exclusively through a competitive auction process, whether for prospecting or mining activities. Additionally, mining leases are issued for a fixed, non-renewable term of 50 years.

- The Mineral Laws (Amendment) Act, 2020 amends the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act) and the Coal Mines (Special Provisions) Act, 2015 (CMSPA). The amendments to the MMDR Act, among other provisions: Allow State governments to begin the auction process for mining leases even before their expiry;
- Provide that approvals, licences, and clearances of the previous lessee are automatically transferred to the new lessee for a period of two years from the date of grant of the new lease; and
- Permit holders of non-exclusive reconnaissance permits to apply for additional licences.

The amendments to the CMSPA are intended to increase coal production and reduce dependence on imports. They allow companies without prior mining experience to participate in coal block auctions and remove end-use restrictions on coal mining. As a result, broader participation is enabled in coal mine auctions for various purposes, including captive use or other uses as specified by the central government.

National Mineral Policy, 2019 replaced the earlier National Mineral Policy, 2008. It was introduced to improve transparency, strengthen enforcement, and promote sustainable mining practices. The policy also seeks to attract private investment and advanced technology by offering incentives such as rights of first refusal in auctions and opportunities for private sector involvement in exploration. It further emphasizes strengthening the regulatory framework through e-governance systems to:

- i. prevents illegal mining and reduces revenue losses;
- ii. ensure complete tracking of mineral movement across the supply chain; and
- iii. enhance awareness and involve local communities in monitoring activities to support enforcement.

The Mineral Conservation and Development (Amendment) Rules, 2018 were introduced to ensure that mineral production continues smoothly despite the expiry of existing mining leases. These rules require general exploration (G2 level) for applicable mining leases (excluding coal, lignite,

and atomic minerals) and set timelines for carrying out exploration activities to enable a seamless transition after lease expiry.

The Mineral (Auction) Amendment Rules, 2017 were notified on 30 November 2017 to accelerate the auction process for major minerals (excluding coal, petroleum, and natural gas). The revised rules introduce flexibility in block allocation during earlier auction rounds instead of delaying until later rounds, which had previously resulted in several blocks being cancelled. They also relax the net-worth requirements for bidders.

Coal Mines (Special Provisions) Act, 2015 (CMSPA) provides a framework for the allocation of coal mines through auctions and allotment. It follows an ascending forward auction system, where bids are based on the price per tonne payable to State governments on coal production. This reform represents a significant change in the coal sector since its nationalisation in 1973 and opens up commercial coal mining to private sector participation.

Pro-Active and Responsive facilitation by Interactive and Virtuous Environmental Single Window Hub (PARIVESH) is an integrated, single-window environmental management system launched by the Ministry of Environment, Forest and Climate Change under the Digital India initiative. It streamlines the process of submitting and tracking applications for various clearances, including environmental, forest, wildlife, and coastal regulation approvals, at both central and State levels.

With the implementation of the National Mineral Policy, 2019 and subsequent amendments to the MMDR Act, India has developed a more transparent, efficient, and investor-friendly framework, creating strong opportunities for investment in the metals and mining sector.

As may be seen, the mining sector in India has potential for further growth due to policy focus on encouraging its growth and also the volume of potential reserves it may mine. The sector also is and can be going forward a major generator of employment. There is also, broadly a dip in the import values, although exports has remain relatively stagnant. Despite the potential it has it its continued expansion, imports of minerals/metals continue to happen in India. These imports happen for those minerals/metals for which there is no domestic reserves, but also happens for those minerals, where India has reserves (in some cases, quite large) such as bauxite, iron ore, coal etc. Given this scenario, it may be of use to understand what may the probable impact, imports of minerals may have on the various aspects of the Indian economy. It would also be of interest to understand what may be the probable impact on the various aspect of the Indian economy, if domestic production of those minerals/metals where India has capacity is increased.

It is with the above context, that the present report has been prepared. The following minerals were identified for the study:

1. Coal
2. Iron Ore
3. Limestone

4. Manganese
5. Bauxite
6. Lead
7. Zinc
8. Copper
9. Tungsten
10. Gold
11. Silver
12. Rock Phosphate
13. Diamond
14. Nickel

The rest of the report is structured as follows: section II covers the trends in imports, section III draws a comparison of India's tariffs on the identified minerals, globally with some of the other top imports, section IV presents some estimates of labor productivity of the identified minerals, section V shows the results of the modeling exercise undertaken showing the possible impact on the economy, if production of minerals/metals and imports of certain metals are increased. Section VI concludes. Appendix I provides an overview of the model used to undertake the modeling activity. Appendix II provides a description of the products, their mapping with harmonized standard codes and their end use categorization. Appendix III provides a description of the production data that was used in the modeling exercise and getting the labor productivity estimates.

CHAPTER | 2

TRENDS IN IMPORTS

The pattern of mineral imports reflects underlying economic conditions, such as gaps in domestic production, technological constraints, and reliance on global supply chains. This chapter examines the changes in India's import trends over time and analyses what they indicate about the country's resource management strategy.

WHAT'S IN THIS CHAPTER:

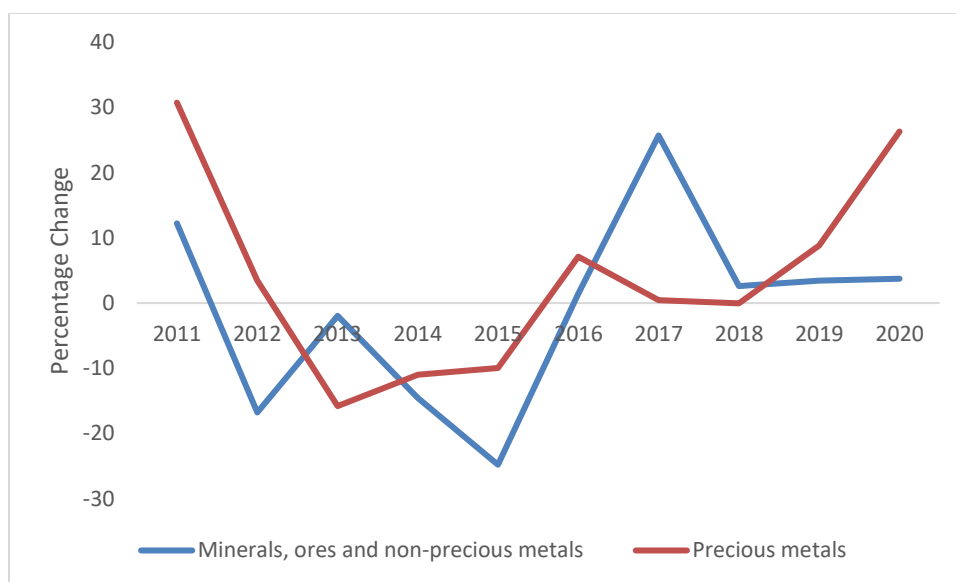
- Global mineral trade patterns
 - Import trends across key minerals
 - Year-wise fluctuations and insights
 - Sector-wise dependency analysis
-

Global Trends

As can be seen in the below figure (figure 4) the global growth rates for Minerals, ores and non-precious metals & Precious metals are quite volatile fluctuating year on year. In the initial years of the period 2011 – 2020, growth rates were negative, with recovery starting only from 2016 onwards. Growth rate for minerals, ores and non-precious metals peaked in 2017, before falling, but not being negative.

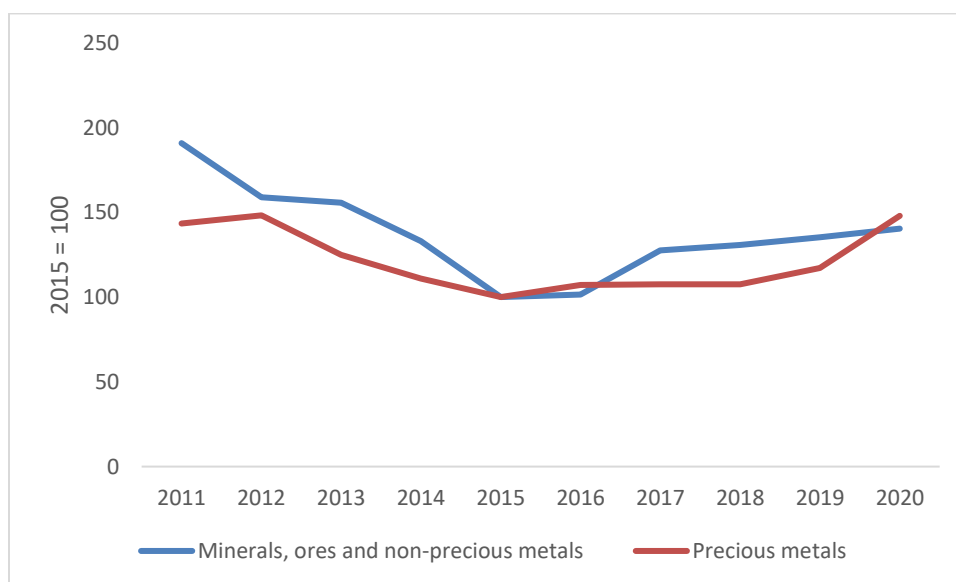
This volatility is also seen, if 2015 is made as base year (as is given in figure 5 below) and the change in growth calculated since then. It can be seen that the index value steadily fell from 2011 to 2015, before slowly recovering. The recovery was more pronounced for minerals, ores and non-precious metals, than for precious metals.

Figure 4: Global growth rates (Minerals, ores and non-precious metals & Precious metals)



Source: UNCTAD

Figure 5: Index of Minerals, ores and non-precious metals & Precious metals (Global, 2015 = 100)



Source: UNCTAD

Iron Ore

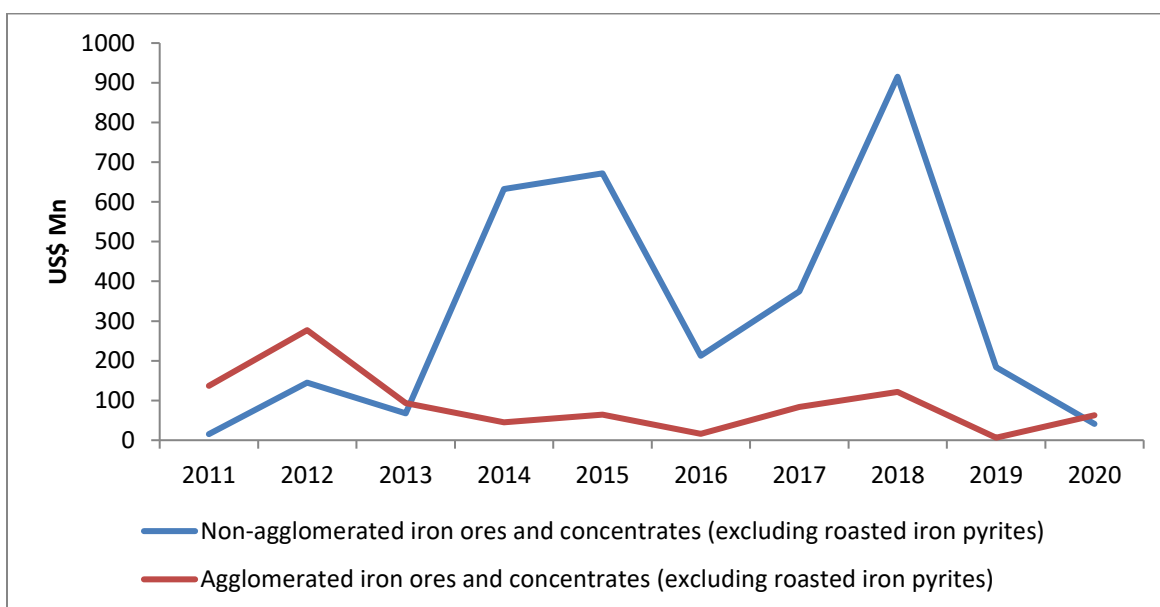


Iron ore is a fundamental resource for industrial production, especially for the steel sector, which plays a vital role in driving infrastructure and manufacturing growth. Despite having substantial reserves, challenges such as variations in ore quality, logistical constraints in transportation, and increasing domestic demand contribute to a complex import scenario. These issues underscore the disconnect between resource availability and its efficient utilization, making iron ore an essential factor in assessing industrial competitiveness.

Iron, with an atomic number of 26, is the most abundant element by mass on Earth. It is primarily used in the production of pig iron, which is further processed to manufacture steel. Iron ore serves as the main raw material for the global iron and steel industry, with nearly 98% of it being utilized for steel production, making it highly significant for manufacturing activities. In the form of steel, iron is widely used in everyday life, including household appliances, infrastructure, transportation, and construction. The entire production of iron ore in India has been reported from the public sector. It contributes around ₹48,107 crore, accounting for about 73% of the total value of principal metallic minerals. In 2019–20, iron ore production reached approximately 246 million tonnes, reflecting a growth of 19%. Odisha is the leading producer, contributing about 59.64% of the total output, followed by Chhattisgarh (14.11%), Karnataka (12.76%), and Jharkhand (10.93%) as of 2020.

In terms of imports of iron ores and concentrates (Figure 6), it is observed that imports under HS 260112 (agglomerated iron ores and concentrates, excluding roasted iron pyrites) were higher during the early years of the 2011–2020 period. However, over time, imports of HS 260111 (non-agglomerated iron ores and concentrates, excluding roasted iron pyrites) gradually increased and significantly surpassed HS 260112 between 2014 and 2018, although the gap narrowed by 2020. The key difference between these categories is that agglomerated ores are formed by combining fine iron particles into larger lumps, whereas non-agglomerated ores consist primarily of fine particles. The rising imports of HS 260111 compared to HS 260112 suggest that importers may prefer to process or agglomerate the ore after import based on their specific requirements. Additionally, despite noticeable fluctuations in the value of imports under both categories, their overall trends remain similar, generally increasing and decreasing in tandem over time.

Figure 6: Imports of Iron Ore (Agglomerated and Non-agglomerated), US\$ Mn



Source: ITC Trade Map

Limestone

Limestone is a key raw material in cement manufacturing and is vital for infrastructure development. With accelerating urbanization and rising construction activities, the demand for limestone has increased substantially. Import patterns indicate that domestic supply often falls short of meeting this demand, highlighting the importance of improving extraction efficiency and strengthening resource management practices.



Limestone is a sedimentary rock mainly composed of calcium carbonate (CaCO_3) and often consists of fossil fragments, shell pieces, and other organic remains. The primary constituents of limestone are calcite and dolomite. It may also contain magnesium carbonate in the form of dolomite $\text{CaMg}(\text{CO}_3)_2$ or magnesite (MgCO_3) along with calcite, and such varieties are referred to as dolomitic or magnesian limestone.

Limestone is an important natural resource obtained from the Earth's crust, mainly through quarrying, where rocks are extracted from large open pits using blasting techniques.

Its usages in sectors/industry include:

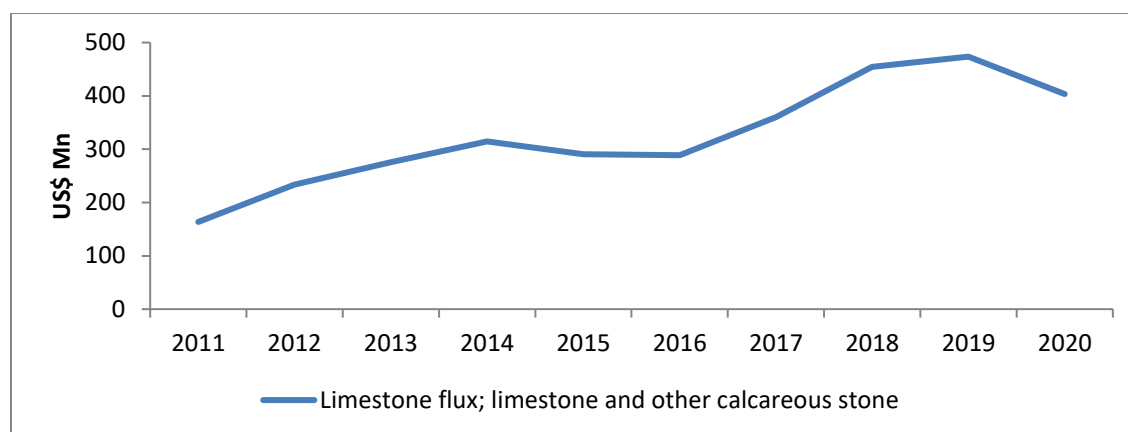
- It is used in cement production by heating powdered limestone with clay. Cement is a key component of mortar and concrete. Mortar is prepared by mixing cement with sand and water, while concrete is made using cement, sand, water, and aggregates such as crushed stone.
- Limestone (calcium carbonate) is a major component in glass manufacturing, contributing significantly to its composition. It helps introduce calcium oxide, which improves durability and chemical resistance, and also acts as a flux.
- It serves as a raw material in the chemical industry for various processes.
- It is used to neutralize acidic soils, thereby improving agricultural productivity. Quicklime and slaked lime are also applied for this purpose, as well as for reducing acidity in water bodies like lakes.

Its different uses include:

- It is widely used as a building material.
- It can be added to food products as a source of calcium for maintaining strong bones and teeth.
- In iron production, limestone helps remove impurities in blast furnaces. Calcium carbonate reacts with silica to form calcium silicate (slag).
- It can be crushed and used in road construction, and slag produced in blast furnaces is also utilized for this purpose.

Turning to imports of limestone, the trend for HS 252100 (limestone flux; limestone and other calcareous stones) is shown in Figure 7. Unlike iron ore, limestone imports do not exhibit large fluctuations in value. Imports were valued at US\$ 162.6 million in 2011 and, except for a decline between 2014 and 2016, showed a steady increase, reaching US\$ 472.65 million in 2019. The decrease observed in 2020 can be attributed to the impact of the COVID-19 pandemic.

Figure 7: Imports of Limestone, US\$ Mn



Source: ITC Trade Map

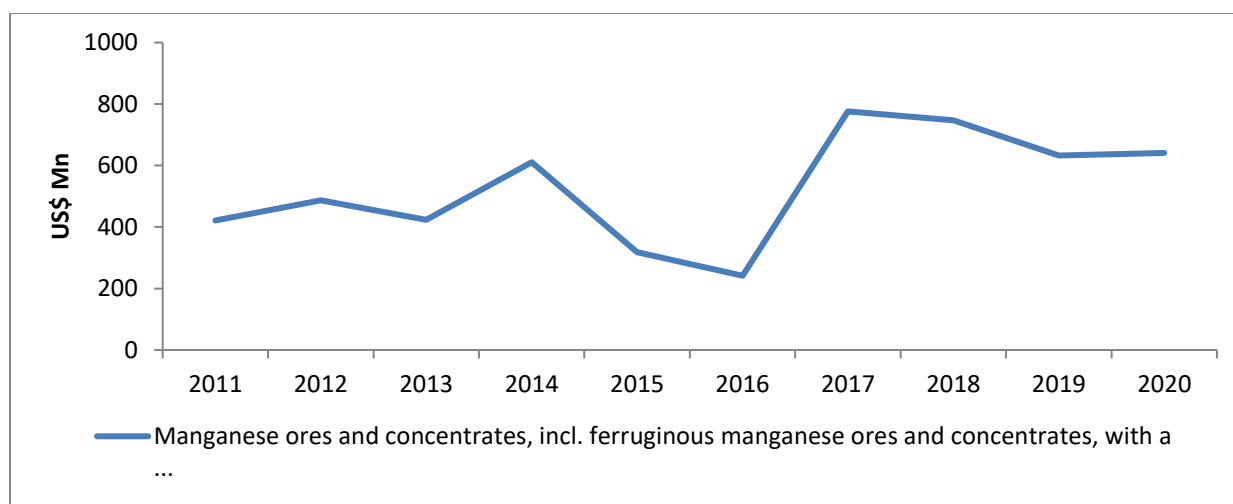
Manganese

Manganese plays a crucial role in steel production, as it improves strength, durability, and resistance to wear. Although India has domestic production, issues such as inconsistent supply and variations in quality make imports necessary to meet industrial demand. The trade patterns of manganese highlight the difficulties in ensuring a stable supply chain within a highly competitive global market.



With regard to manganese imports (figure 8), import of manganese ore and concentrate happens under HS 260200 (Manganese ores and concentrates, incl. ferruginous manganese ores and concentrates..) and unlike imports of limestone, is subjected to frequent changes in its import values. Imports of manganese ore stood at US\$ 421.50 million in 2011, and despite the large changes in its import value, reached its highest level (in the period 201 – 2020) in 2017, when US\$ 776.22 million of it was imported. It fell thereafter and in 2020, US\$ 641.01 million was imported.

Figure 8: Imports of Manganese Ore, US\$ Mn



Source: ITC Trade Map

Aluminum (Bauxite)

Bauxite is the main source of aluminum, a metal extensively used in sectors such as transportation, construction, and packaging.

Although India has substantial reserves of bauxite, constraints in refining capacity and processing efficiency affect its import patterns. This underscores the need to strengthen the entire value chain, from extraction to final production.



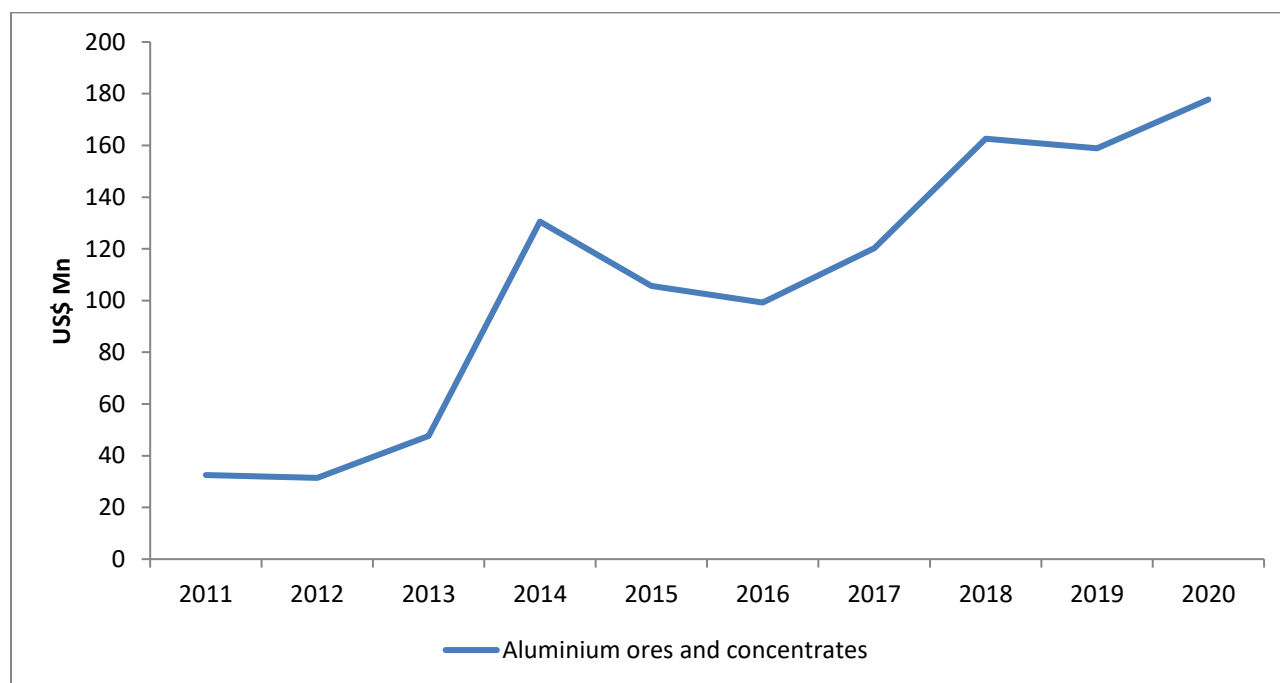
Bauxite is the primary and only ore of aluminum. It has wide-ranging applications across metallurgy, chemical industries, and as a raw material in construction and road aggregates. It is the most important material for producing aluminum metal and is also used in industries such as chemicals, refractory bricks, abrasives, cement, steel, and petroleum. Lateritic bauxite is commonly utilized as a building material. Calcined bauxite, produced by heating high-alumina bauxite in rotary, round, or shaft kilns at high temperatures, is used as an anti-skid aggregate in road construction to enhance safety in accident-prone areas. Odisha emerged as the leading producer, accounting for about 65% of total production during 2018–19.

Bauxite is mainly used to produce alumina through the Bayer process. The aluminum industry typically requires bauxite with a minimum of 40% Al_2O_3 content, although lower-grade ores may

also be used when blended appropriately, depending on properties such as solubility in caustic soda and low silica content. The Bureau of Indian Standards (BIS) has established specifications under IS:5953-1985 (reaffirmed in 2008 and 2014) for metallurgical-grade bauxite. In the steel industry, bauxite is used as a slag corrector in place of fluorite, and generally bauxite containing 45–54% Al_2O_3 and a maximum of 5% SiO_2 is preferred.

India also imports aluminum ores and concentrates (figure 4), despite higher reserves of the same and an up-stream aluminum smelting industry with captive mines. Imports of aluminum ores and concentrates (figure 9) happens under HS 260600 (Aluminum ores and concentrates). While it is not subjected to the large variations as imports of manganese and imports of iron ore, imports of aluminum have also in a few years witnessed a sudden increase in its value. For example, imports rose from US\$ 47.66 million in 2013 to US\$ 130.54 million in 2014, before falling to US\$ 105.70 million in 2015. Imports again rapidly rose from US\$ 120.22 million in 2017 to US\$ 162.63 million in 2018 and it has continued its upward trend standing at US\$ 177.76 million in 2020.

Figure 9: Imports of Aluminum ores and concentrates, US\$ Mn



Source: ITC Trade Map

Lead

Lead is a strategically significant metal, mainly used in the production of batteries, particularly for automotive and energy storage applications. Its demand is closely associated with industrial development and technological progress. Import patterns are shaped by factors such as recycling capacity, environmental regulations, and the growing expansion of energy storage systems.



Lead is a soft, dense metal with a silvery-white or greyish appearance. It is highly malleable and ductile, but a poor conductor of electricity.

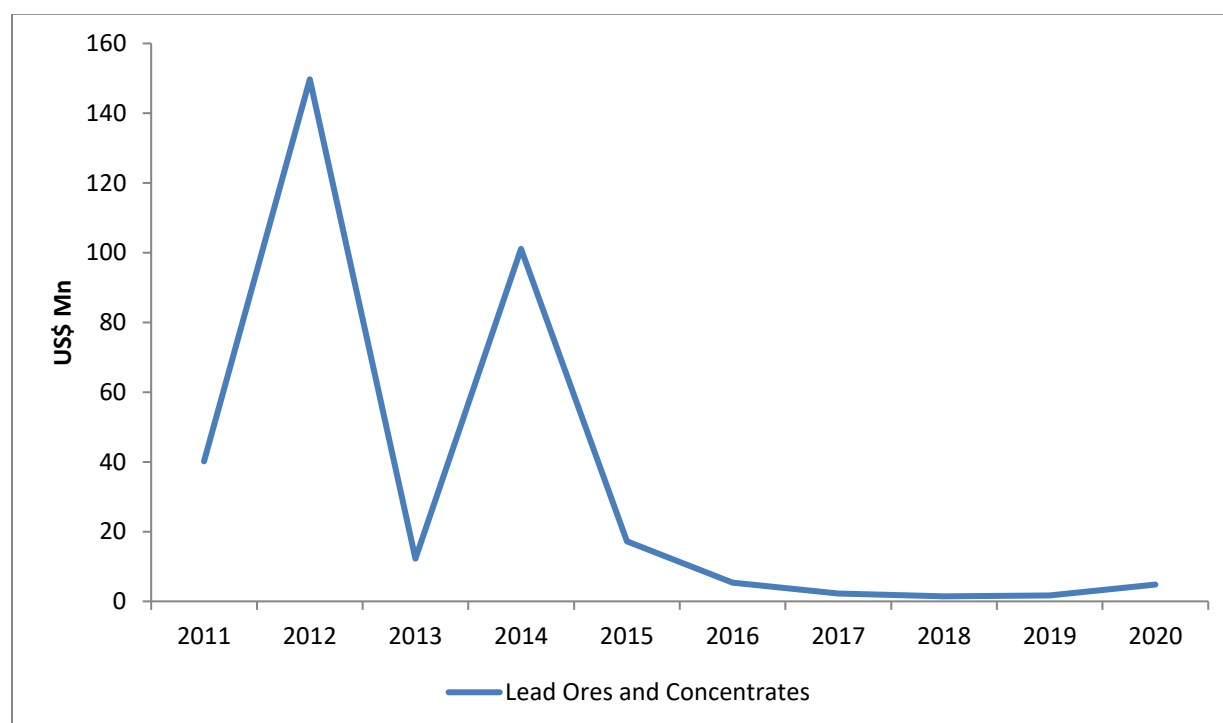
Lead has a wide range of applications, including use in automobile batteries, pigments, ammunition, cable sheathing, lifting weights, diving belts, lead crystal glass, radiation shielding, and certain types of solder. When used in batteries, cable coverings, and radiation protection, lead can be fully recycled without losing its properties. As a result, a well-established lead recycling industry exists in the country. Globally, major lead reserves are found in countries such as Australia, China, Russia, Peru, the USA, and Mexico.

Lead mining is largely carried out in countries like Australia, Bolivia, Canada, China, and India. In India, primary lead production stood at 1,32,316 tonnes in 2019–20, showing a significant decline of 33% compared to the previous year. Rajasthan was the only state reporting production of lead concentrate during this period, and Hindustan Zinc Limited (HZL) remains the sole producer of primary lead in the country.

The growing demand for lead, particularly from the lead-acid battery sector, is largely met through the recycling of lead scrap. However, due to health risks associated with recycling, operations are regulated through licensing by the Central Pollution Control Board. It is estimated that more than 50% of refined lead globally is obtained from recycled sources, and this method requires only about one-third of the energy compared to extraction from ores. In the international market, the price of lead was around \$1999.68 per tonne, slightly higher than \$1965.47 per tonne, while the average domestic price during 2019–20 was approximately ₹149.26 per kg.

With respect to imports of lead ores and concentrates (figure 10), unlike the previous minerals, its imports have followed a decreased trend, preceded by large increases and fall in import value for the period 2011 – 2020. Imports of HS 260700 (Lead Ores and Concentrates) stood at US\$ 40.17 million in 2011, before rising to US\$ 149.76 million in 2012. It then fell to US\$ 12.30 million in 2013, before rising to US\$ 101.10 million in 2014, falling again to US\$ 17.20 million in 2015 and then again falling to US\$ 5.36 million in 2016. Imports of HS 260700 have stayed in single digits since then.

Figure 10: Imports of Lead ores and concentrates, US\$ Mn



Source: ITC Trade Map

Zinc

Zinc plays an important role in protecting steel through galvanization, which improves durability and resistance to corrosion. Although India has strengthened its domestic production over time, imports are still required due to increasing industrial demand. These trends indicate a balance between growing self-sufficiency and changing consumption requirements.



Zinc is a moderately brittle metal at room temperature and has a silvery-grey appearance when oxidation is removed. It is primarily used for galvanizing other metals, especially iron, to protect them from corrosion. Galvanized steel is widely used in applications such as automobile bodies, street lighting poles, safety barriers, and suspension bridges. A significant amount of zinc is also utilized in the production of die-cast components, which are essential in the automobile, electrical, and hardware industries. In addition, zinc is used in various alloys, including brass, nickel silver, and aluminum solder.

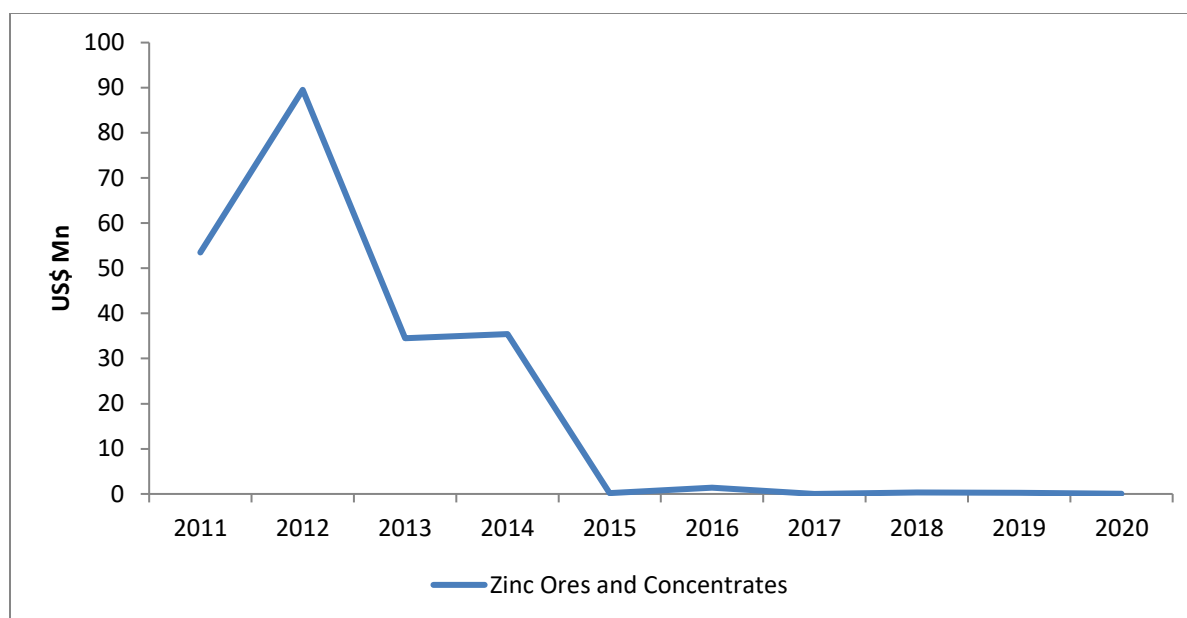
Zinc oxide is extensively used in the manufacture of a wide range of products, including paints, rubber, cosmetics, pharmaceuticals, plastics, inks, soaps, batteries, textiles, and electrical equipment. Zinc sulphide is commonly used in the production of luminous paints, fluorescent lamps, and X-ray screens.

The production of zinc ingot metal at 6,96,283 tonnes in 2018-19 decreased by 12% from 7,91,461 tonnes in the previous year. Hindustan Zinc Ltd, contributed 100% of the total output. In the financial year 2018, the volume of zinc production in India amounted to approximately 1.54 million metric tons. This was estimated to slightly increase to around 1.66 million metric tons in the fiscal year 2021¹.

As with lead, imports of zinc ores and concentrates also follow a rapidly decreasing trend for the period 2011 – 2020 (figure 11). Imports of zinc ores and concentrers fall under HS 260800. Imports rose from US\$ 53.51 million in 2011 to US\$ 89.53 million in 2012. It fell to US\$ 34.48 million in 2013 and from 2015 onwards, imports are barely happening.

¹ https://ibm.gov.in/writereaddata/files/10142020122001Lead_Zinc_2019_AR.pdf

Figure 11: Imports of Zinc ores and concentrates, US\$ Mn



Source: ITC Trade Map

Copper

Copper is essential for electrical systems, renewable energy infrastructure, and telecommunications. Due to limited availability of high-grade ore and constraints in refining capacity, India depends on imports to meet its requirements.

This highlights the need to

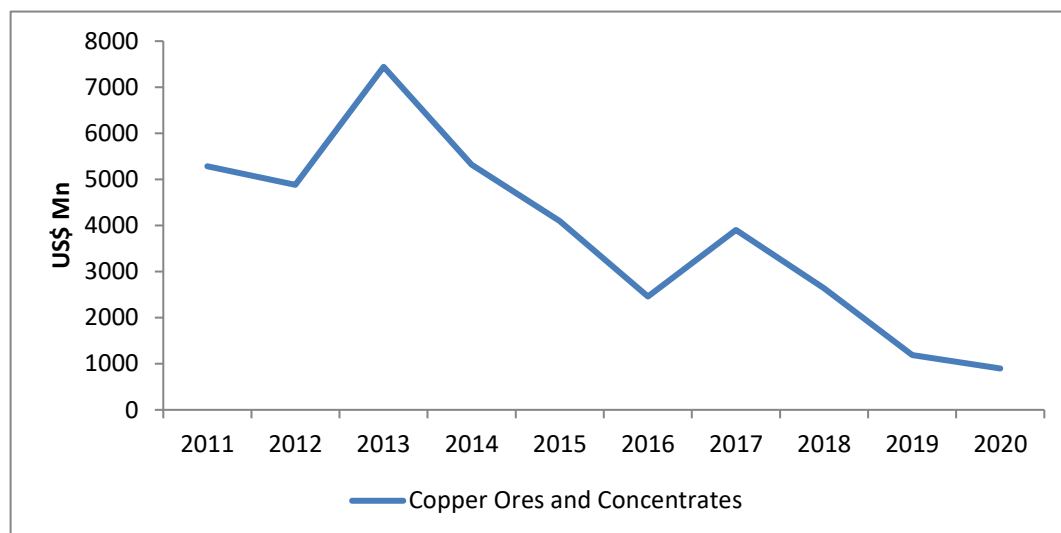
strengthen domestic capabilities to support increasing technological and infrastructure demands.



Although India has a well-established copper smelting and upstream industry, most of its copper ore and concentrate needs are fulfilled via imports (figure 12). Imports of copper ores and concentrates happen under HS 260300. As with lead and zinc (both of which are non-ferrous

metals), imports of copper has also followed a mostly decreasing trend in terms of import value. Imports of HS 260300 peaked at US\$ 7443.38 million in 2013, within the period 2011 – 2020, thereafter following a decreasing trend till 2020.

Figure 12: Imports of Copper ores and concentrates, US\$ Mn



Source: ITC Trade Map

Tungsten

Tungsten is a scarce and high-value mineral widely used in defense, aerospace, and advanced manufacturing because of its superior strength and high heat resistance. India's limited domestic reserves result in a strong dependence on imports, underscoring its strategic significance for national security and industrial development.

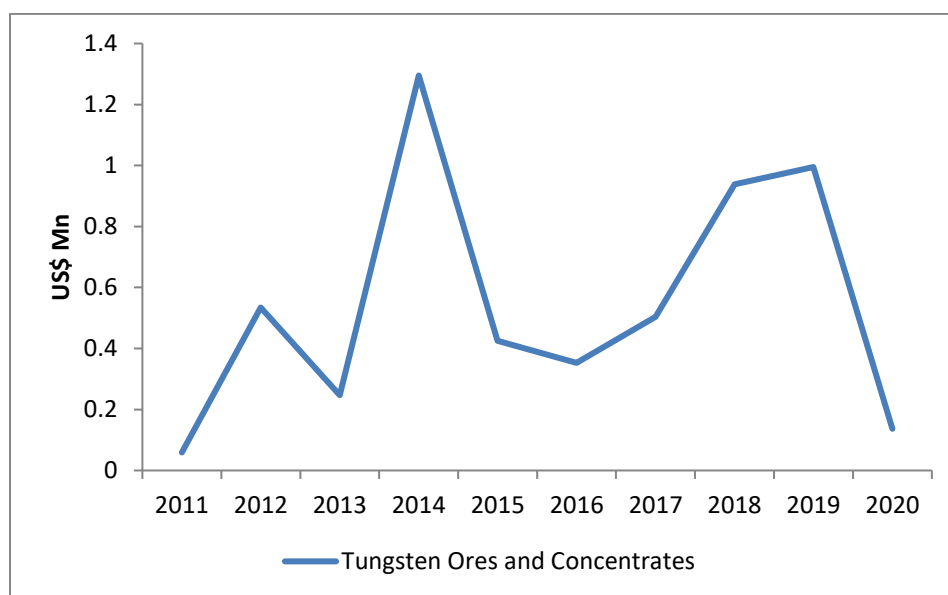


Tungsten is a rare metal that occurs naturally mainly in combination with other elements rather than in pure form. It is widely used in the production of ballistic missiles and drilling tools due to its exceptional hardness. Tungsten alloys are also used in making light bulb filaments, X-ray tubes, and materials for radiation shielding. Owing to its extremely high boiling point, tungsten is highly

suitable for high-temperature applications, including arc welding electrodes and heating elements that operate at very high temperatures.

Imports of HS 261100 (Tungsten Ores and Concentrates) are given in figure 13. It is immediately seen that India hardly imports tungsten in any significant quantities at all over the period 2011 – 2020, with the imports rising above US\$ one million for a single year (2014).

Figure 13: Imports of Tungsten ores and concentrates, US\$ Mn



Source: ITC Trade Map

Gold

Gold occupies a distinctive role in India's economy, blending cultural importance with financial significance. It is extensively used in jewellery and also serves as a reliable store of value. Import trends are shaped by consumer preferences, investment behaviour, and fluctuations in global prices.



Gold is one of the least reactive chemical elements and remains solid under standard conditions. In its pure form, it appears as a bright, slightly reddish-yellow metal and is known for being dense,

soft, highly malleable, and ductile. Due to its versatility, gold is used in applications such as mining, electroplating, and the formation of amalgam alloys. It is also highly resistant to corrosion, which enhances its durability. Gold plays an important role in the aerospace industry, where reliability and efficiency are critical. It is widely used in the electronics sector, particularly in devices such as mobile phones, calculators, and televisions, owing to its excellent conductivity and connectivity. In addition, certain gold compounds are used in medicine, where gold salts serve as anti-inflammatory agents.

The number of gold mines recorded in India is 5 in 2019-20 as compared to 6 mines in the preceding year. Beginning from 2009-10, the production, export and import has been constantly rising. During 2018-19, the production of gold ore was 565.65 thousand tonnes which increased by 5% in 2019-20 which was, 591.25 thousand tonnes². The major downfall in the export of gold was between 2018-19 and 2019-20. On the other hand the imports didn't experience a downfall. The domestic demand for gold is majorly met through imports in the country.

Karnataka was the only state producing primary gold in 2019–20. Gold, along with other precious metals such as silver and selenium, is also recovered during the process of copper refining at the plant located in Dahej, Bharuch district of Gujarat. The largest reserves of primary gold ore are found in Bihar (44%), followed by Rajasthan (25%), Karnataka (21%), West Bengal and Andhra Pradesh (3% each), and Jharkhand (2%).

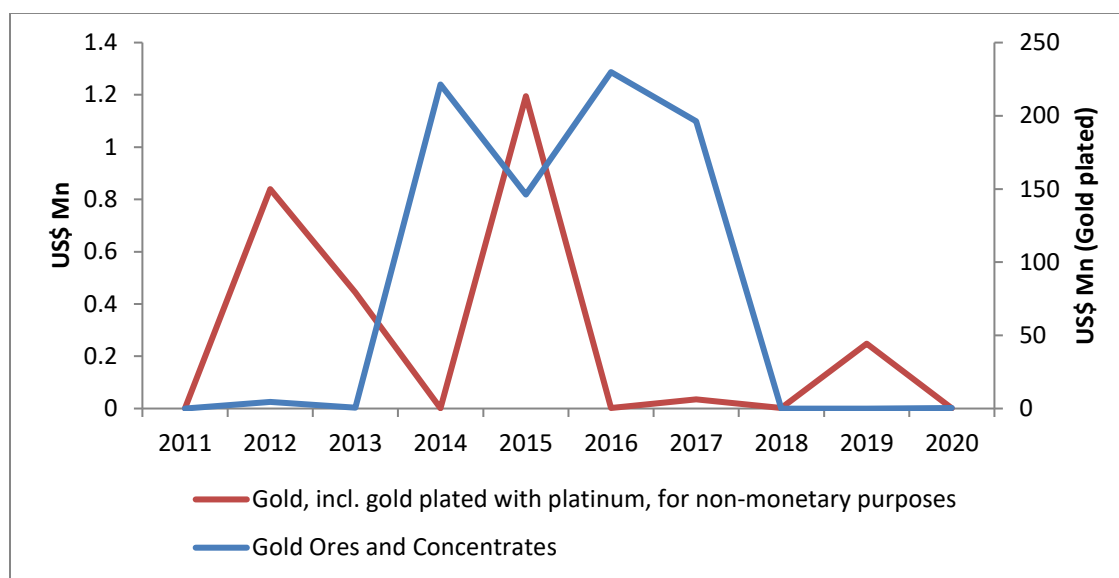
In India, gold bullion is available in both primary and secondary forms. The total production of gold bullion during 2019–20 was 8,364 kg, which shows a sharp decline of 34% compared to 12,614 kg in 2018–19.

With respect to gold, unlike the above minerals, imports happen under ores and concentrates (figure 14) and also under semi-processed gold (figure 15).

Imports of gold ores and concentrates (HS 26190) have been constant at US\$ 4.49 million for the period 2012 to 2020. Imports of gold, for non-monetary purpose (HS 710811) are quite small for the period 2011 to 2020. It only crossed US\$ one million for only one year (2015). Most of India's gold imports (figure 15) happen under HS 710812 (Gold, incl. gold plated with platinum, unwrought, for non-monetary purposes (excluding gold) and HS 710813 (Gold, incl. gold plated with platinum, in semi-manufactured forms, for non-monetary purposes). Although initially, imports are happening under both the tariff sub-headings (HS 710812 and HS 710813), over the period 2014 to 2020, a majority of the imports are happening under HS 710812. This may be as gold under this tariff heading comes in unwrought form and is therefore further processed for domestic value addition. Imports under both the tariff sub-headings have been on a decreasing trend, in terms of value.

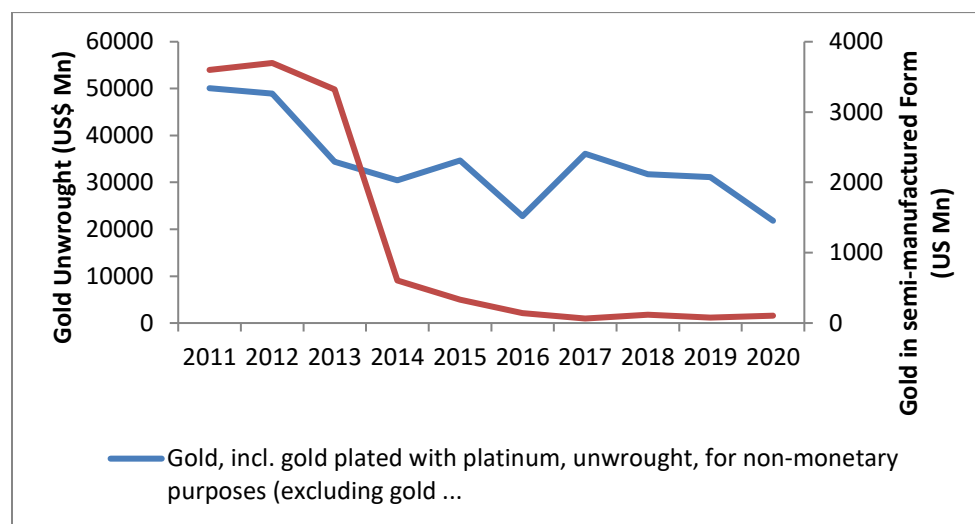
² https://ibm.gov.in/writereaddata/files/01282022115038Gold_2020.pdf (Page 2)

Figure 14: Imports of Gold ores and concentrates and Gold (non-monetary purpose), US\$ Mn



Source: ITC Trade Map

Figure 15: Imports of Gold (unwrought and semi-processed), US\$ Mn



Source: ITC Trade Map

Silver

Silver has both industrial and investment significance, with uses spanning from electronics to renewable energy technologies. Its demand has been increasing due to technological progress, while import patterns highlight its dual role as both a commodity and a precious metal.



Silver is a shiny, soft, white metal known for its high lustre. It is both a precious and industrial metal and possesses the highest electrical and thermal conductivity among all metals. Represented by the symbol Ag, silver has a relatively high melting point of 962°C. Although it is not highly reactive, it tends to tarnish over time due to exposure to hydrogen sulphide in the air. It is also less dense than gold. Silver is generally obtained as a by-product during the smelting and refining of lead, zinc, and copper ores. India has very limited native silver deposits, with only a few small occurrences found in Rajasthan. As of 1 April 2015, the total silver ore resources in the country were estimated at 511,954,635 tonnes.

Due to its highest electrical conductivity properties, Silver is used in all-electric appliances, electric circuits, coating for electronic conductors. It is also used in jewellery and ornaments. Today Silver is invaluable to solder and brazing alloys, batteries, dentistry, glass coatings, LED chips, medicine, nuclear reactors, and photography. The manufacturing sector is the biggest consumer of Silver; industries manufacturing electronics use Silver due to its properties. Besides these, demand in the jewellery and pharma sector is also increasing. It is also forecasted that increase in the use of solar energy will expand markets for Silver. In India, total reserves/resources of Silver in the country as of 1.4.2015, 7,172 tonnes are under 'Reserves'.³

As with gold, imports of silver also happens in both ores and concentrates (figure 16) and in unwrought and semi-processed form (figure 12=7)

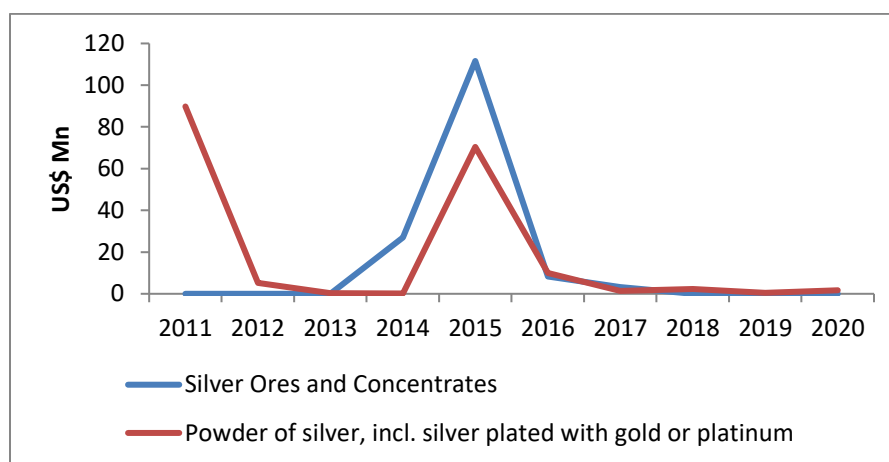
Imports of silver ores and concentrates happens under HS 261610 (Silver Ores and Concentrates) and imports of silver powder happens under HS 7100610 (Powder of silver, incl. silver plated with gold or platinum). Unlike gold, imports of silver ores have seen a lot of fluctuations, peaking at US\$ 111.600 million in 2015. Thereafter, it fell and hardly anything was imported between 2018

³ https://ibm.gov.in/writereaddata/files/09072021154419Silver_2020.pdf; page number 3

to 2020. Imports of silver powder also follow a similar pattern with respect to silver ore, albeit at lower import values. Imports of silver powder also in 2015 (US\$ 70.43 million).

Most silver imports into India fall under HS 710691 (unwrought silver, including silver plated with gold or platinum, excluding powder form) and HS 710692 (semi-manufactured silver, including silver plated with gold or platinum). Between these two categories, imports under HS 710692 generally remained higher than those under HS 710691 throughout the period, except in 2011 and 2020. This pattern differs from gold imports, where unwrought forms typically accounted for a larger share compared to semi-processed gold.

Figure 16: Imports of Silver Ore and Silver Powder, US\$ Mn



Source: ITC Trade Map

Figure 17: Imports of Silver (unwrought and semi-processed), US\$ Mn



Source: ITC Trade Map

Rock Phosphate

Rock phosphate is a crucial raw material for fertilizer production, playing a direct role in agricultural productivity and food security. Due to limited domestic reserves, India relies heavily on imports, highlighting the importance of maintaining stable supply chains to support the agricultural sector.

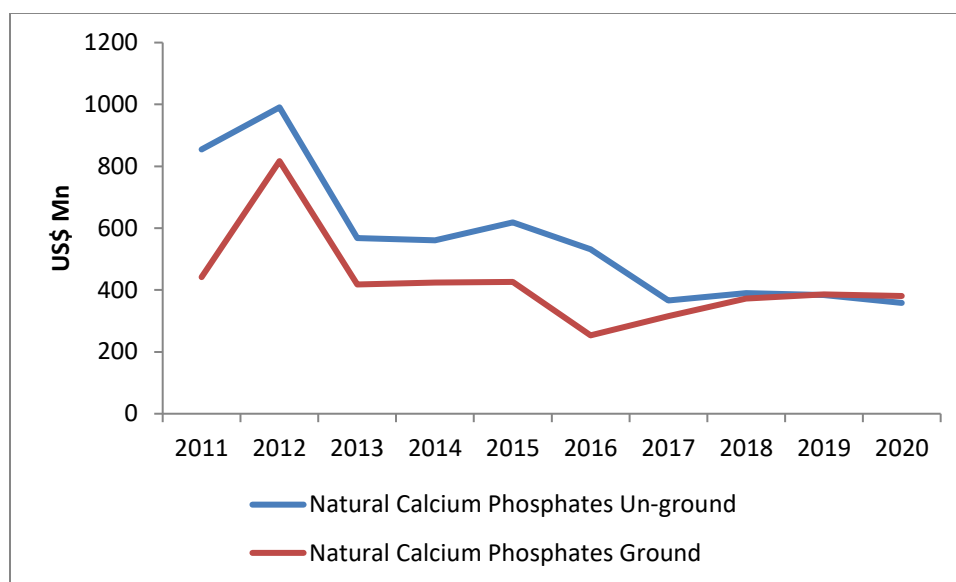


Phosphate rock, also known as phosphorite, refers to any rock that contains a high concentration of phosphorus. The most economical and significant source of phosphorus is obtained by mining and processing phosphate rock from deposits found across the world. Some phosphate rock is used in the production of calcium phosphate supplements for animal nutrition, while pure phosphorus is utilized in manufacturing various industrial chemicals. However, its most important application is in the production of phosphate fertilizers used in agriculture. Fertilizers are commonly rated as “N-P-K,” where phosphorus represents the “P” component and plays a vital role in plant growth, particularly in enabling plants to capture solar energy and initiate photosynthesis.

In India, phosphate rock production is mainly concentrated in two states—Rajasthan and Madhya Pradesh. The country currently has around 30 lakh metric tonnes of phosphorite reserves, and efforts are being made by the government to increase production. These deposits are located in regions such as Rajasthan, central peninsular India, Hirapur (Madhya Pradesh), Lalitpur (Uttar Pradesh), the Mussoorie syncline, and the Cuddapah basin in Andhra Pradesh.

Imports of rock phosphate (figure 18) happen under two tariff sub-headings HS 251010 (Natural Calcium Phosphates Un-ground) and HS 251020 (Natural Calcium Phosphates Ground). Like the import trends for many of the minerals given below, there is a clear discernable trend with respect to imports under calcium phosphates. Between 2011 to 2017 imports under HS 251010 exceeded imports of HS 251020. From 2018 onwards, the gap between the two narrowed and imports under both HS 251010 and HS 251020 happen at similar import volumes.

Figure 18: Imports of Calcium Phosphates, US\$ Mn



Source: ITC Trade Map

Diamond

India is a leading global hub for diamond cutting and polishing, even though it has limited natural reserves. The country imports rough diamonds and enhances their value through processing, positioning itself as a major player in the global gemstone industry.

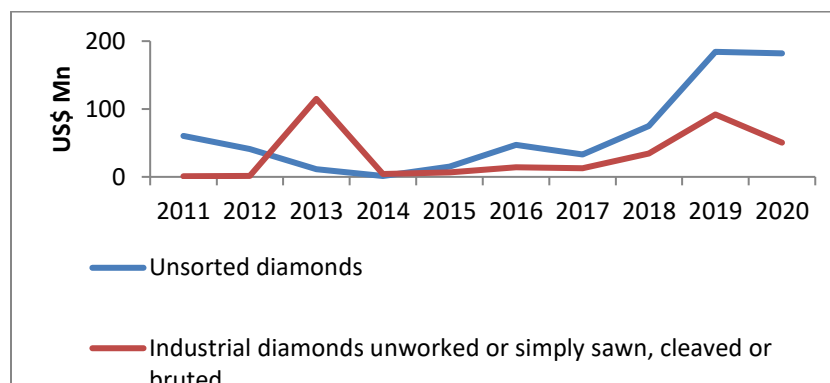


Like with gold and silver imports of diamonds also happen under different product categories (figure 19 and figure 20).

In figure 19, imports under HS 71020 (Unsorted diamonds) and HS 710221 (Industrial diamonds unworked or simply sawn, cleaved or bruted) are given. Imports of HS 710221 reached a high of US\$ 114.94 million in 2013, falling sharply thereafter before again picking up. From the period 2015 onwards, imports under HS 710220 were more than HS 710221, with the difference between import values slowly increasing over the years.

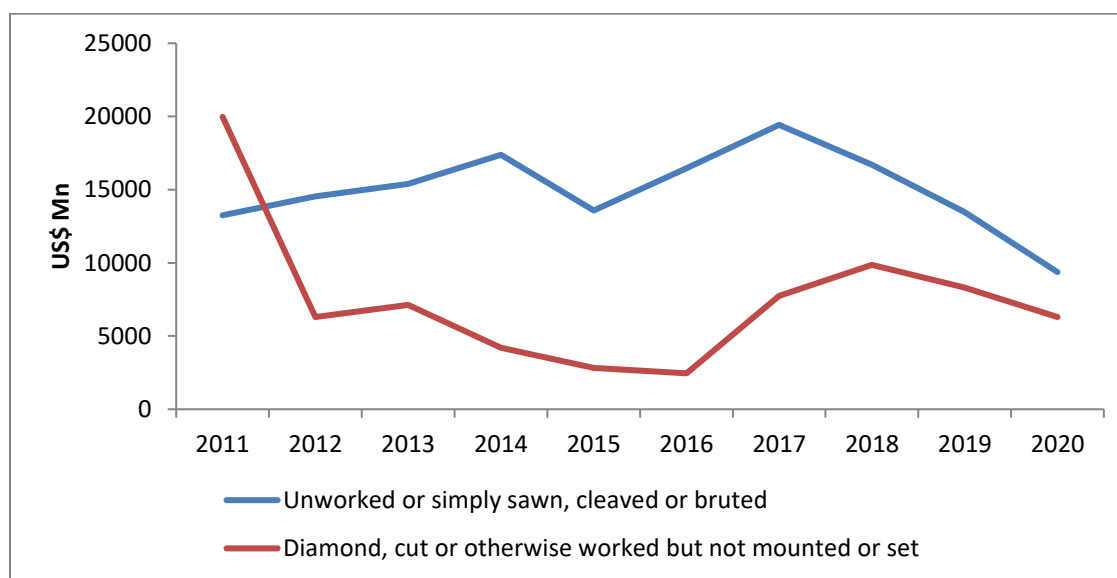
India has over the years built its diamond jewelry business, including processing raw diamonds. This may explain the large volumes of imports under HS 710231 (Unworked or simply sawn, cleaved or bruted) and HS 710239 ("Diamond, cut or otherwise worked but not mounted or set") as given in figure 20. Between these two tariff sub-headings, a large part of the imports is happening under HS 710231, indicating that there diamonds are possibly being imported for further value addition either for domestic consumption or exports.

Figure 19: Imports of Unsorted and industrial diamonds, US\$ Mn



Source: ITC Trade Map

Figure 20: Imports of Unworked and non-mounted diamond, US\$ Mn



Source: ITC Trade Map

Nickel

Nickel has become increasingly significant in modern industries, especially in the production of stainless steel and in battery technologies used for electric vehicles. Growing demand, combined with limited domestic availability, has resulted in higher imports, highlighting its importance in the shift toward advanced manufacturing and clean energy systems.



Nickel is a shiny, silvery-white metal known for its durability and ability to easily form alloys with other metals. It is widely used across various sectors, including stainless steel production, catalysis in chemical industries, electroplating, heat-resistant alloys, non-ferrous metal alloying, as well as in space, defence, and rocket applications. It is also used in nickel-cadmium batteries.

Nickel is found in many industrial and consumer products such as stainless steel, alnico magnets, coins, filters, binders, rechargeable batteries, foundry products, electric guitar strings, microphone components, and specialized alloys. It is also used for plating purposes and as a coloring agent to provide a green tint in glass.

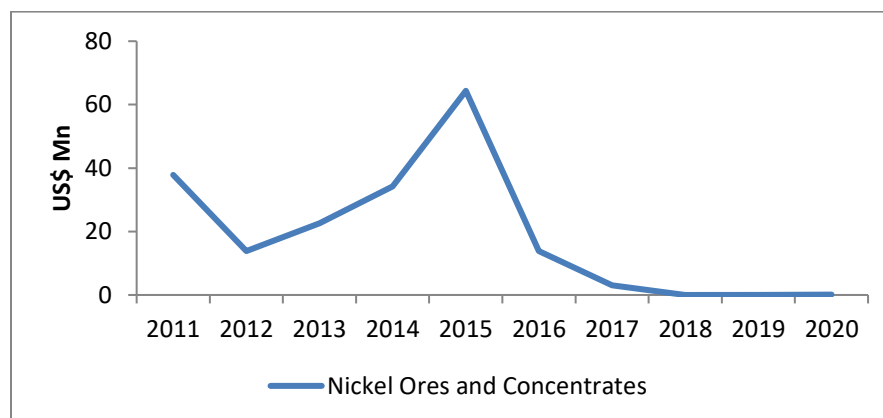
Primarily an alloy metal, nickel is extensively used in nickel steel and nickel cast iron, along with a variety of other alloys such as nickel bronze, brass, and combinations with copper, chromium, aluminium, lead, cobalt, silver, and gold.

Nickel also acts as an important catalyst in several industrial processes, including hydrogenation of vegetable oils, hydrocarbon reforming, and the production of fertilizers, pesticides, and fungicides. Nickel sulphate is a key compound used in electroplating, enamelling processes, preparation of nickel-based compounds, and catalytic applications. Nickel-based alloys, especially stainless steel with higher nickel content, are used in demanding environments such as gas turbines and chemical plants.

India has traditionally been dependent on imports for nickel. However, Hindustan Copper Limited (HCL) has established the country's first nickel production facility at the Indian Copper Complex (ICC) in Ghatsila, Jharkhand. This "Nickel, Copper and Acid Recovery Plant" is the first in India to produce nickel metal of London Metal Exchange (LME) grade from primary resources.

Imports of nickel ores and concentrated (figure 21) happen via HS 260400 (Nickel Ores and Concentrates). Between 2011 to 2014, imports of nickel rose from US\$ 13.885 million to US\$ 64.36 million. Thereafter it fell and in terms of value, its imports are that not much.

Figure 21: Imports of Nickel ores and concentrates, US\$ Mn



Source: ITC Trade Map

Coal

Coal continues to be a major source of energy in India, playing a key role in power generation and industrial operations. Although the country has substantial domestic

reserves, issues such as supply inefficiencies and variations in coal quality necessitate ongoing imports, highlighting the gap between resource availability and actual energy demand.

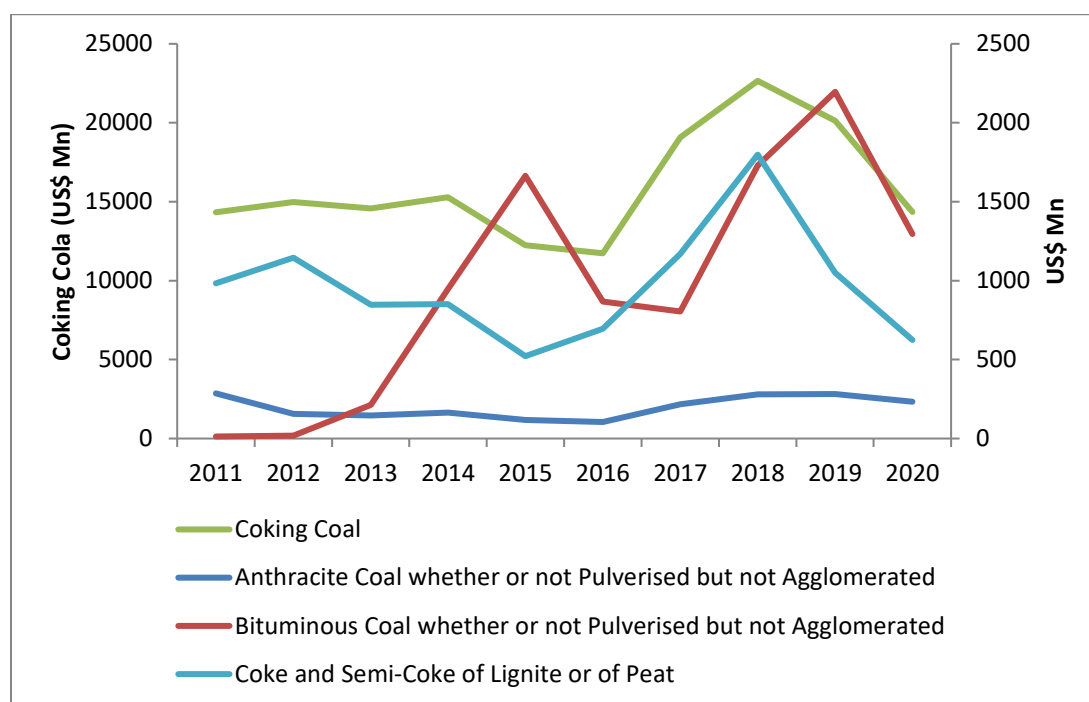


India is the second-largest producer of coal and holds the fourth-largest coal reserves in the world. Coal India Limited (CIL) is the leading producer in the country. In addition to CIL, coal mining is carried out by The Singareni Collieries Company Limited (SCCL) and several public and private sector companies that have been allocated coal blocks under the CM(SP) Act, 2015 and the MMDR Act, 1957. The total coal production in India during FY 2020 was approximately 716.29 million tonnes (provisional). During the same period, around 69% of coking coal imports came from Australia, about 59% of non-coking coal imports were sourced from Indonesia, and nearly 22% were imported from South Africa.

Coal imports (Figure 22) are categorized under four tariff sub-headings: HS 270111 (anthracite coal, whether or not pulverized but not agglomerated), HS 270112 (bituminous coal, whether or

not pulverized but not agglomerated), HS 270119 (coking coal), and HS 270400 (coke and semi-coke of lignite or peat). Among these, coking coal accounts for the highest share of imports across all years, primarily due to its significant use in the iron and steel industry. Although the import values vary over time, all four categories generally exhibit similar trends, suggesting common underlying supply and demand dynamics.

Figure 22: Imports of Coal, US\$ Mn



Source: ITC Trade Map

In the above figures, the import values for different metals and minerals selected for the study were given. It was seen that across all the metals and minerals, import values fluctuated on a regular basis, driven by domestic demand supply movements and global commodity price movements. To better understand the growth of imports, the compounded annual growth rate (CAGR) for the different minerals and metals identified for the study are given (table 3). As with the import values, across many of the identified products, there exists a great difference in CAGR in the whole period (2011 – 2020), first half period (2011 – 2015) and second half period (2016 – 2020). Imports of some of the products grew over the entire period, but exhibited negative growth across one period, which was offset by higher growth in the other period (iron ore, manganese, tungsten). Some of the products exhibited positive CAGR across all the three periods (limestone, bauxite, bituminous coal) and some of the products exhibited negative growth across all the periods.

Table 3: Compounded Annual Growth Rates for Selected Minerals and Metals (percentage)

Mineral	HS Code	Product Description	Full Period CAGR	2011 - 2015 CAGR	2016 - 2020 CAGR
Iron Ore	260111	Non-agglomerated iron ores and concentrates (excluding roasted iron pyrites)	11.47%	157.41%	-33.85%
	260112	Agglomerated iron ores and concentrates (excluding roasted iron pyrites)	-8.29%	-17.01%	40.99%
Limestone	252100	Limestone flux; limestone and other calcareous stone	10.55%	15.45%	8.72%
Manganese	260200	Manganese ores and concentrates, incl. ferruginous manganese ores and concentrates, with a ...	4.77%	-6.78%	27.64%
Bauxite	260600	Aluminium ores and concentrates	20.75%	34.22%	15.68%
Lead	260700	Lead Ores and Concentrates	-20.92%	-19.11%	-2.39%
Zinc	260800	Zinc Ores and Concentrates	-58.17%	-76.09%	-65.01%
Copper	260300	Copper Ores and Concentrates	-17.88%	-6.17%	-22.26%
Tungsten	261100	Tungsten Ores and Concentrates	9.81%	63.83%	-21.07%
Gold	261690	Gold Ores and Concentrates	-	-	-80.84%
	710811	Gold, incl. gold plated with platinum, for non-monetary purposes	0.00%	487.95%	-15.91%
	710812	Gold, incl. gold plated with platinum, unwrought, for non-monetary purposes (excluding gold ...	-8.82%	-8.78%	-1.10%
	710813	Gold, incl. gold plated with platinum, in semi-manufactured forms, for non-monetary purposes	-32.55%	-44.85%	-7.31%
Silver	261610	Silver Ores and Concentrates	-	-	-100.00%
	710610	Powder of silver, incl. silver plated with gold or platinum	-35.80%	-5.89%	-36.02%
	710691	Silver, incl. silver plated with gold or platinum, unwrought (excluding silver in powder form)	-23.67%	-1.32%	-31.26%

	710692	Silver, incl. silver plated with gold or platinum, semi-manufactured	-1.63%	-28.14%	45.95%
Rock Phosphate	251010	Natural Calcium Phosphates Un-ground	-9.21%	-7.74%	-9.38%
	251020	Natural Calcium Phosphates Ground	-1.66%	-0.93%	10.68%
Diamond	710210	Unsorted diamonds	13.00%	-29.57%	40.08%
	710221	Industrial diamonds unworked or simply sawn, cleaved or bruted	55.67%	63.09%	37.67%
	710231	Unworked or simply sawn, cleaved or bruted	-3.78%	0.60%	-13.13%
	710239	Diamond, cut or otherwise worked but not mounted or set	-12.04%	-38.68%	26.57%
Nickel	260400	Nickel Ores and Concentrates	-45.53%	11.20%	-63.74%
Coal	270111	Anthracite Coal whether or not Pulverised but not Agglomerated	-49.09%	14.19%	-71.87%
	270112	Bituminous Coal whether or not Pulverised but not Agglomerated	-2.23%	-19.80%	22.18%
	270119	Coking Coal	66.82%	236.64%	10.53%
	270400	Coke and Semi-Coke of Lignite or of Peat	0.02%	-3.85%	5.15%

Source: Authors calculations based on ITC Trade Map data

CHAPTER | 3

COMPARISON OF IMPORT TARIFFS



Tariff policies serve as important tools that shape trade flows and affect domestic competitiveness. This chapter examines India's tariff structure in comparison with that of global economies, providing insights into policy alignment and its strategic position in international markets.

WHAT'S IN THIS CHAPTER:

- India's tariff structure overview
- Comparison with major economies
- Duty variations across minerals

To better understand the structure of import duties of the minerals/metals identified here for India, a comparison is made with some major importers of minerals. The same is presented in table below:

It can quite clearly be seen that not only does India have the highest import tariff (duty) in more or less each tariff sub-heading, it also has the least number of tariff sub-headings under zero duty (13.79 percent).

As calculated from the below table (table 4), Canada and European Union have an import duty of zero for all the tariff sub-headings. Many of the other countries, i.e. Japan (0.06 percent), South Korea (1.40 percent), Chinese Taipei (0.02 percent), United States of America (0.22 percent), and Malaysia (0.17 percent) have very low average import duties across all the tariff sub-headings mentioned below. The average import duty of China across all the tariff sub-headings is also quite low at 1.33 percent. Incidentally, China is also the largest importer of minerals and metal ores.

India's average import duty across all the tariff sub-headings mentioned below stands at 4.91 percent with the range varying from 2.5 percent for copper ores and concentrates to 12.5 percent for the precious metals and stones.

Table 4: Average Import Duty for Certain Tariff Subheadings for Select Countries (Percentage)

[illegible]

710811	Gold, incl. gold plated with platinum, for non-monetary purposes	0	0	0	0	3	0	0	0.00	12.5
710812	Gold, incl. gold plated with platinum, unwrought, for non-monetary purposes (excluding gold ...	0	0	0	0	3	0	0	2.05	12.5
710813	Gold, incl. gold plated with platinum, in semi-manufactured forms, for non-monetary purposes	0	0	0	0	3	0	0	1.4	12.5
261610	Silver Ores and Concentrates	0	0	0	0	0	0	0	0.00	2.5
710610	Powder of silver, incl. silver plated with gold or platinum	0	0	0	0	3	0	0	0.00	12.5
710691	Silver, incl. silver plated with gold or platinum, unwrought (excluding silver in powder form)	0	0	0	0	3	0	0	1.50	12.5
710692	Silver, incl. silver plated with gold or platinum, semi-manufactured	0	0	0	0	3	0	0	1.50	12.5
251010	Natural Calcium Phosphates Un-ground	0	3	0	0	0.5	0	0	0.00	5

251020	Natural Calcium Phosphates Ground	0	3	0	0	3	0	0	0.00	5
710210	Unsorted diamonds	0	3	0	0	1	0	0	0.00	0
710221	Industrial diamonds unworked or simply sawn, cleaved or bruted	0	0	0	0	1	0	0	0.00	0
710229	Crushed diamonds	0	0	0	0	5	0	0	0.00	7.5
710231	Unworked or simply sawn, cleaved or bruted	0	3	0	0	1	0	0	0.00	0
710239	Diamond, cut or otherwise worked but not mounted or set	0	4	0	0	5	0	0	0.00	7.5
260400	Nickel Ores and Concentrates	0	0	0	0	0	0	0	0.00	0
270111	Anthracite Coal whether or not Pulverised but not Agglomerated	0	3	0	0	0	0	0	0.00	2.5
270112	Bituminous Coal whether or not Pulverised but not Agglomerated	0	4.5	0	0	0	0	0	0.00	2.5
270119	Coking Coal	0	5	0	0	0	0	0	0.00	2.5
270400	Coke and Semi-Coke of Lignite or of Peat	0	5	0	1.6	3	0.5	0	0.00	5

Source: WTO

CHAPTER | 4

LABOR PRODUCTIVITY

Efficiency in the mining sector depends not only on output levels but also on workforce productivity and the adoption of advanced technologies. This chapter analyzes productivity trends and highlights the key factors that influence operational performance in the sector.

WHAT'S IN THIS CHAPTER:

- Productivity trends in mining sector
- Comparison with overall industries
- Efficiency gaps and challenges
- Impact on growth and employment

Productivity is also an important driver for growth and trends in labor productivity can also help us sometimes develop a narrative on how the industry is faring vis-à-vis the rest of the economy. Total sales (reported in INR million) and total number of employees were collected for companies engaged in mining of the above minerals/metals. Labor productivity measure was arrived at by dividing total sales by the total number of employees. In this sense, higher the number that would mean higher the labor productivity, lower the number, means lower productivity, i.e. the average contribution per employee is decreasing. The labor productivity figures are listed in table 5

As a measure of comparison, the above data was also collected for all companies and all mining companies. The data on sales and employees were collected from the companies listed in the prowess database of CMIE. The labor productivity figures for all companies and all mining companies are given in table 6.

With respect to iron ore (table 5), it is observed that, one there is a large amount of variation in the entire period 2011 – 2021. But while there is this large variation, with a few rises in productivity, overall, the trend has been of a decreasing one. The measure of labor productivity stood at 21.13 in 2011 and in 2021, it stood at 6.40.

Turning to limestone (table 5), a somewhat similar trend is noticed in terms of variation. But, unlike iron ore, there is no persistent long-term decreasing trend. Rather the taper in labor

productivity towards the end of the period is sudden. For example, in 2011, labor productivity stood at 17.64, thereafter decreased to 16.93 and then increased to 26.63. It stayed in the 12 to 25 range thereafter, fluctuating year on year, before falling rapidly to 4.90 in 2021.

Manganese on the other hand (table 5) follows a persistent downward trend for the entire period, with it being noticeable from 2016 onwards. With respect to bauxite, the divergence in the productivity figures do not vary by that much year on year. Except for a few years (including 2016, where it dropped to 8.89), the labor productivity figures stayed in the range of 10 to 13 (exceeding in some years beyond that).

In Lead & Zinc (table 5), one problem that is being faced in terms of understanding the long term labor productivity trends is the number of missing variables on account of missing data. But, from what has been calculated, it can be seen that it has amongst the highest labor productivity numbers, although the divergence in the figures as was seen above, is also present here. Copper also has the same problem as Lead & Zinc in terms of missing variables. Copper also has the lowest labor productivity numbers. The issue of missing variables. Figures for tungsten & nickel could not be calculated on account of lack of data on number of employees.

With respect to gold & silver (table 5), it is seen that there exists comparatively stable labor productivity figures with not much variation across most years. The same to a certain degree can also be said with respect to diamond (table 3) which is a precious stone/metal in which Indian industry has developed competence in. Excepting for a few years (including 2020, where it dropped from 21.33 in 2019 to 9.60), the figures with respect to diamond are also relatively stable. Unlike the metals/minerals which were mentioned above, with respect to coal (table 5), there has been a steady rise in the labor productivity figures from 2011 till 2021. Particularly since 2014, the labor productivity numbers for coal have been steadily rising.

How do the figures for the minerals/metals used in the study compare to the labor productivity figures for all industries and for all mining industries? To compare them, the figures for labor productivity for all industries and for all mining industries is presented in table 6. With respect to productivity figure for all industries, it can be seen that vis-à-vis most of the minerals/metal mentioned above, it exhibits considerably less volatility and is mostly stable around a range. This is probably expected as these are figures for all industries and any volatility in a certain segment may be adjusted with relatively less volatility in other segments. With respect to labor productivity for all mining industry (table 6), it also exhibits considerable less divergence in its figures as compared to most of the minerals/metals mentioned above. The reason for the same may be similar as that of productivity figures for all industries.

In order to better understand what the labor productivity figures are on average for all the industries mentioned above and if the divergences are also reflected in terms of figures, the mean and standard deviation for all the industries mentioned above are given in table 7. As can be seen, across most industries the standard deviation is high.

Table 5: Labor Productivity of Mining Companies

Year	Iron	Limestone	Manganese	Bauxite	Lead & Zinc	Copper	Gold & Silver	Diamond	Coal
2011	21.13	17.64	13.23	13.74	-	1.49	-	57.96	5.77
2012	12.74	16.93	14.95	17.03	-	1.81	26.15	57.32	5.56
2013	5.68	26.63	17.04	18.83	-	1.83	19.81	57.71	5.77
2014	13.66	19.32	20.83	12.64	37.89	2.06	19.57	48.00	5.87
2015	17.21	13.73	15.46	14.48	44.66	2.14	16.65	46.79	6.43
2016	13.42	13.26	9.58	8.89	41.31	2.63	17.30	55.34	7.77
2017	13.93	16.83	8.75	12.92	74.65	3.80	20.04	48.93	9.68
2018	11.11	25.37	7.59	14.24	-	-	21.24	45.03	12.39
2019	8.83	12.27	8.20	12.89	-	-	20.12	40.45	11.49
2020	6.61	11.75	7.10	11.05	80.55	-	21.12	21.33	53.48
2021	6.40	4.90	5.10	10.27	80.25	-	19.26	9.60	37.91

Source: Authors calculations based on Prowess data

Table 6: Labor Productivity of All Companies and All Mining Companies

	All Companies	All Mining Companies
2011	13.07	23.29
2012	12.33	23.55
2013	15.32	19.04
2014	14.43	25.50
2015	14.25	26.94
2016	13.78	24.29
2017	14.38	23.18
2018	16.10	28.95
2019	15.24	24.69
2020	14.90	18.58
2021	11.16	12.17

Source: Authors calculations based on Prowess data

Table 7: Labor Productivity Average and Standard Deviation (All Companies, All Mining Companies, Mining Company data for selected minerals)

	Average	Standard Deviation
All Companies	14.09	1.37
All Mining Companies	22.74	4.41
Iron	11.88	4.58
Limestone	16.24	0.89
Manganese	11.62	4.73
Bauxite	13.36	2.72
Lead & Zinc*	59.88	17.40
Copper**	2.25	0.71
Tungsten & Nickel	0.12	0.39
Gold & Silver	20.12	2.44
Diamond	44.41	14.91
Coal	14.74	15.14

Source: Authors calculations based on Prowess data

calculated for the period 2014 – 2017 & 2020 – 2021 on account of missing values

calculated for the period 2011 – 2017 on account of missing values

CHAPTER | 5

ECONOMIC MODELLING RESULTS

Policy choices in the mineral sector have significant macroeconomic implications. This chapter employs simulation-based analysis to measure how changes in imports and production affect key economic indicators, including GDP, employment, and the trade balance.

WHAT'S IN THIS CHAPTER:

- Simulation-based policy analysis
- Impact on GDP, trade, and prices
- Employment and wage effects
- Scenario-based economic outcomes

To understand the impact trade may possibly have on the mining industry, the economy as a whole and also employment, a series of simulations were conducted. These simulations were done based on the Global Trade Analysis Project Model (GTAP) and were done broadly on the following basis: first the impact of a one percent increase in each mineral's import may have (where demand is met through imports) was modelled and the results given in tables 8 - 15; second, a scenario where domestic production may increase by one percent and imports of those minerals where domestic production does not happen may also increase by one percent was modelled (tables 16 - 18). These scenarios were undertaken to understand better the cross sectoral linkages and the possible impact on the economy as a whole.

Simulation on Imports of Copper, Tungsten and Nickel

The first scenarios, examine, what may happen if imports may increase by one percent for each mineral separately (copper, tungsten and nickel) and the results are explained. The areas which were examined for explanation are – production, market prices, exports, imports, labor demand, wages, impact on real GDP, aggregate exports and imports. The tables may be read as thus: for tables concerning production, market prices, exports, and import, the columns are the simulations for each mineral. For example, a one percent increase in copper falls under the heading copper, a one percent increase in tungsten falls under the heading tungsten. The rows have the industries, which have been impacted with the changes being reflected in percentage

terms. The impact on the manufacturing industry and industries relating to oil, coal, and gas are examined given the cross sectoral linkages between them.

Production

Turning now to copper (column 2), it may be seen that the impact that though the possible magnitude of impact may be small, it may still be happening in varying degrees, across mining and manufacturing including a probable fall in copper production. Why is the possible magnitude of impact so small? A reason for this may be that, most of India's copper ores and concentrates is imported with little contribution from domestic production, a one percent increase in the imports of something which is not available domestically, may not impact the other industries studied here, as a majority of the raw material is already imported. The noticeable fall in copper ore and concentrate production may again be due to the sensitivity of movement of commodity prices to trade trends. A rise in imports, presumably maybe through a fall in tariffs, may possibly lead to a reduction in domestic production. But since domestic production contributes a small percentage to overall consumption, with the majority being met by imports, the rise in imports may not significantly affect other industries. In fact, a situation may be envisaged where a fall in tariffs, may lead to pass through of lower costs which may help in increasing domestic production of value-added copper products for use in downstream industries. Some of this may also be seen in both the production of other minerals/ores and also the rise in production of those industries which utilizes copper more intensely than others. The possible rise in production of the other minerals may be, as there is a fall in the importing cost of copper, and this additional costs savings may be percolated across the mineral/commodity industry, possibly resulting in additional production. It may be seen that, some of the manufacturing industries may be experiencing a possible increase in production, vis-à-vis others. Particularly, some of the labor-intensive industries such as textiles, apparel manufacturing may possibly experience a decline in production, whereas others such as non-ferrous metal production, manufacturing of machinery, metal products, gas manufacturing may possibly see an increase in production.

The possible impact of nickel (column 3), and tungsten (column 4) simulations may not be much. While noting the importance of these minerals, a reason for this may be the much lower import volumes of these minerals (as was seen in the import trends) and their general usage in the manufacturing industry vis-à-vis other minerals like iron ore or bauxite.

In summary, it may be noted that, while the possible magnitude of impact may be small, the cross sectoral linkages in production are explored and it is seen that while the production of the mineral whose imports have increased in a simulation has decreased, this has been offset by an increase in the production of other minerals. Further, those manufacturing industries which use these minerals more intensively than others have experienced a positive shock. Part of the reason for the same may also be like in the movements in market prices and labor wages which are covered in the following sections.

Market Prices

For copper (table 9, column 2), it may be seen that, although the magnitude of the proposed impact may be small on market prices, it may be seen that there may be an overall decrease in prices for minerals and a general overall increase in prices of manufacturing products. The possible general decrease in prices of minerals, may be due to an increased production of the minerals being studied and given the sensitivities commodities are with respect to production and prices, the possible increase in production may have led to a decrease in prices. With respect to the manufacturing industry, it may be seen that bearing non-ferrous metal manufacturing and other manufacturing, there may probably be an overall increase in prices. A part of the reason for this may be the probable reallocation of labor, as to possibly attract more labor in the industry to help in the possible expansion of production, higher wages possibly be offered, nudging an overall increase in prices across the board. The decrease in prices in non-ferrous metal production may be the direct pass through of lower copper prices in the industry.

With respect to tungsten (table 9, column 3) and nickel (table 9, column 4) it may be seen that there hardly may be any change in price. Part of the reason may be that the import volumes may not be that large to begin with or their share in the manufacturing industry is limited vis-à-vis some of the other minerals like bauxite, coal, iron ore etc.

Exports and Imports

As with production and market prices, it may be seen that the possible impact on exports (table 10) across the manufacturing and mining industry, with respect to a possible one percent increase in imports of copper, nickel and tungsten may be small. At least with respect to copper, it may be seen that, although the magnitude may be small, there may possibly be a negative impact on exports. A part of this may be due to (given the relatively fixed nature of labor supply in the short run) an increase in the wages of the workers being employed. This may in turn probably lead to higher consumption, which in turn may lead to producers catering to domestic demand, instead of exports. Also, with respect to tungsten and nickel, as was mentioned before, given the comparative small space they occupy, the increase in imports may have an even lesser impact than copper.

As is well known, India is a consumption driven economy, even though policy initiatives have been taken to boost manufacturing. Despite this, the size of the manufacturing industry continues to remain small, when compared to services. This may imply that, despite a possible increase in the base and production of the manufacturing industry, some part of the domestic demand may be met through imports. This may also be true if there is a continued rise in wages. With respect to imports (table 11), the above explanation may be the reason, what may be driving higher imports, in case of the copper imports simulation. However, as may be seen, the magnitude of the impact continues to be small, because, as mentioned before, most of domestic consumption of copper ores/concentrates is being met through imports. The impact of tungsten and nickel continues to be of a smaller magnitude than copper.

Table 8: Change in Production due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

Product	Copper	Nickel	Tungsten
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>	<i>Column 4</i>
Coal: mining and agglomeration of hard coal, lignite and peat	-0.00013	0	0
Oil: extraction of crude petroleum, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00028	0	0
Gas: extraction of natural gas, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00036	0	0
Iron ore	0.00123	0	0
Limestone	0.00117	0	0
Manganese	0.00113	0	0
Bauxite	0.00116	0	0
Lead	0.00096	0	0
Zinc	0.00097	0	0
Copper	-0.58116	0	0
Tungsten	0.00084	0	-0.01
Gold	0.00124	0	0
Silver	0.00121	0	0
Red Phosphorus	0.00083	0	0
Diamond	0.00123	0	0
Nickel	0.00084	-0.01	0
Manufacture of textiles	-0.00065	0	0

Manufacture of wearing apparel	-0.00142	0	0
Manufacture of leather and related products	-0.00156	0	0
Lumber: manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	0.00041	0	0
Paper & Paper Products: includes printing and reproduction of recorded media	-0.00046	0	0
Petroleum & Coke: manufacture of coke and refined petroleum products	0.00009	0	0
Manufacture of chemicals and chemical products	-0.00068	0	0
Manufacture of pharmaceuticals, medicinal chemical and botanical products	-0.00156	0	0
Manufacture of rubber and plastics products	-0.00014	0	0
Manufacture of other non-	0.00116	0	0

metallic mineral products			
Iron & Steel: basic production and casting	0.00054	0	0
Non-Ferrous Metals: production and casting of copper, aluminium, zinc, lead, gold, and silver	0.00284	0	0
Manufacture of fabricated metal products, except machinery and equipment	0.0008	0	0
Manufacture of computer, electronic and optical products	-0.00036	0	0
Manufacture of electrical equipment	0	0	0
Manufacture of machinery and equipment n.e.c.	0.00032	0	0
Manufacture of motor vehicles, trailers and semi-trailers	0.00045	0	0
Manufacture of other transport equipment	-0.00062	0	0
Other Manufacturing: includes furniture	0.00108	0	0
Electricity; steam and air conditioning supply	0.00018	0	0

Gas manufacture, distribution	0.00006	0	0
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Table 9: Change in Market Prices, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

Product	Copper	Nickel	Tungsten
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>	<i>Column 4</i>
Coal: mining and agglomeration of hard coal, lignite and peat	0.00042	0	0
Oil: extraction of crude petroleum, service activities incidental to oil and gas extraction excluding surveying (part)	0.00009	0	0
Gas: extraction of natural gas, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00002	0	0
Iron ore	-0.00085	0	0
Limestone	-0.00086	0	0
Manganese	-0.00087	0	0
Bauxite	-0.00086	0	0
Lead	-0.00092	0	0
Zinc	-0.00091	0	0
Copper	-0.22949	0	0
Tungsten	-0.00097	0	0
Gold	-0.00085	0	0
Silver	-0.00085	0	0
Red Phosphorus	-0.00097	0	0
Diamond	-0.00085	0	0
Nickel	-0.00097	0	0
Manufacture of textiles	0.00052	0	0
Manufacture of wearing apparel	0.00055	0	0
Manufacture of leather and related products	0.00055	0	0
Lumber: manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	0.00064	0	0

Paper & Paper Products: includes printing and reproduction of recorded media	0.00051	0	0
Petroleum & Coke: manufacture of coke and refined petroleum products	0.00015	0	0
Manufacture of chemicals and chemical products	0.0003	0	0
Manufacture of pharmaceuticals, medicinal chemical and botanical products	0.00057	0	0
Manufacture of rubber and plastics products	0.00046	0	0
Manufacture of other non-metallic mineral products	0.00009	0	0
Iron & Steel: basic production and casting	0.00021	0	0
Non-Ferrous Metals: production and casting of copper, aluminium, zinc, lead, gold, and silver	-0.00055	0	0
Manufacture of fabricated metal products, except machinery and equipment	0.00011	0	0
Manufacture of computer, electronic and optical products	0.00029	0	0
Manufacture of electrical equipment	0.00029	0	0
Manufacture of machinery and equipment n.e.c.	0.00033	0	0
Manufacture of motor vehicles, trailers and semi-trailers	0.00036	0	0
Manufacture of other transport equipment	0.00037	0	0
Other Manufacturing: includes furniture	-0.00016	0	0
Electricity; steam and air conditioning supply	0.00047	0	0
Gas manufacture, distribution	0.00077	0	0

Table 10: Change in Exports, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

Product	Copper	Nickel	Tungsten
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>	<i>Column 4</i>
Coal: mining and agglomeration of hard coal, lignite and peat	-0.00071	0	0
Oil: extraction of crude petroleum, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00088	0	0
Gas: extraction of natural gas, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00009	0	0
Iron ore	0.00126	0	0
Limestone	0.00127	0	0
Manganese	0.00129	0	0
Bauxite	0.00127	0	0
Lead	0.00137	0	0
Zinc	0.00136	0	0
Copper	0.37121	0	0
Tungsten	0.00145	0	0.01
Gold	0.00126	0	0
Silver	0.00126	0	0
Red Phosphorus	0.00146	0	0
Diamond	0.00126	0	0
Nickel	0.00145	0.01	0
Manufacture of textiles	-0.00359	0	0
Manufacture of wearing apparel	-0.0037	0	0
Manufacture of leather and related products	-0.00441	0	0
Lumber: manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	-0.0041	0	0
Paper & Paper Products: includes printing and reproduction of recorded media	-0.00267	0	0

Petroleum & Coke: manufacture of coke and refined petroleum products	-0.00053	0	0
Manufacture of chemicals and chemical products	-0.00195	0	0
Manufacture of pharmaceuticals, medicinal chemical and botanical products	-0.00354	0	0
Manufacture of rubber and plastics products	-0.00289	0	0
Manufacture of other non-metallic mineral products	-0.00052	0	0
Iron & Steel: basic production and casting	-0.00119	0	0
Non-Ferrous Metals: production and casting of copper, aluminium, zinc, lead, gold, and silver	0.00397	0	0
Manufacture of fabricated metal products, except machinery and equipment	-0.00093	0	0
Manufacture of computer, electronic and optical products	-0.00281	0	0
Manufacture of electrical equipment	-0.00268	0	0
Manufacture of machinery and equipment n.e.c.	-0.00271	0	0
Manufacture of motor vehicles, trailers and semi-trailers	-0.00199	0	0
Manufacture of other transport equipment	-0.00309	0	0
Other Manufacturing: includes furniture	0.00082	0	0
Electricity; steam and air conditioning supply	-0.00269	0	0
Gas manufacture, distribution	-0.00437	0	0

Table 11: Change in Imports, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

Product	Copper	Nickel	Tungsten
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>	<i>Column 4</i>
Coal: mining and agglomeration of hard coal, lignite and peat	0.00084	0	0
Oil: extraction of crude petroleum, service activities incidental to oil and gas extraction excluding surveying (part)	0.00016	0	0
Gas: extraction of natural gas, service activities incidental to oil and gas extraction excluding surveying (part)	-0.00025	0	0
Iron ore	0.00073	0	0
Limestone	0.00068	0	0
Manganese	0.00064	0	0
Bauxite	0.00068	0	0
Lead	0.00036	0	0
Zinc	0.00039	0	0
Copper	1	0	0
Tungsten	0.00001	0	1
Gold	0.00074	0	0
Silver	0.00072	0	0
Red Phosphorus	-0.00002	0	0
Diamond	0.00074	0	0
Nickel	0.00001	1	0
Manufacture of textiles	0.00183	0	0
Manufacture of wearing apparel	0.0022	0	0
Manufacture of leather and related products	0.00175	0	0
Lumber: manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	0.00274	0	0
Paper & Paper Products: includes printing and reproduction of recorded media	0.00119	0	0

Petroleum & Coke: manufacture of coke and refined petroleum products	0.00055	0	0
Manufacture of chemicals and chemical products	0.00038	0	0
Manufacture of pharmaceuticals, medicinal chemical and botanical products	0.00077	0	0
Manufacture of rubber and plastics products	0.00154	0	0
Manufacture of other non-metallic mineral products	0.00143	0	0
Iron & Steel: basic production and casting	0.00147	0	0
Non-Ferrous Metals: production and casting of copper, aluminium, zinc, lead, gold, and silver	0.0001	0	0
Manufacture of fabricated metal products, except machinery and equipment	0.00143	0	0
Manufacture of computer, electronic and optical products	0.00128	0	0
Manufacture of electrical equipment	0.0018	0	0
Manufacture of machinery and equipment n.e.c.	0.00205	0	0
Manufacture of motor vehicles, trailers and semi-trailers	0.00195	0	0
Manufacture of other transport equipment	0.0007	0	0
Other Manufacturing: includes furniture	0.00078	0	0
Electricity; steam and air conditioning supply	0.00148	0	0
Gas manufacture, distribution	0.00208	0	0

Table 12: Change in Tariff to facilitate one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

Mineral	Tariff Change
Copper	-3.24
Nickel	-1.13
Tungsten	-1.13

It is seen in this table that it is changes in tariffs that is driving the growth of imports. While, across all the simulations, there has been only a one percent increase in imports of each mineral, the change in tariffs is however not uniform. There is a wide variation in the change in tariffs as is seen in the table. This divergence may be explained by the differing nature of demand for each mineral in the Indian economy and their specific supply side nature.

Table 13: Change in Employment, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

	Clerks	Service Shop/Factory Workers	Technical/ Professional workers	Officers/Managerial Professionals	Agricultural and Other Low Skilled Workers
Copper	0.00033	0.0005	0.0001	0.00043	0.00045
Nickel	0	0	0	0	0
Tungsten	0	0	0	0	0

Table 14: Change in Wages, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

	Clerks	Service Shop/Factory Workers	Technical/ Professional workers	Officers/Managerial Professionals	Agricultural and Other Low Skilled Workers
Copper	0.00055	0.00055	0.00055	0.00055	0.00055
Nickel	0	0	0	0	0
Tungsten	0	0	0	0	0

As it may be observed, the possible impact of a rise in one percent imports of copper, nickel and tungsten may possibly have a small impact on employment and wages. But, this impact, despite possibly being small, may be positive especially in the case of copper as is seen.

Table 15: Change in real GDP, Aggregate Exports and Imports, due to one percent rise in imports of Copper, Nickel and Tungsten (Percentage Change)

	Real GDP (% change)	% Change (US\$ Mn)	Aggregate Exports (% change)	% Change (US\$ Mn)	Aggregate Imports (% change)	% Change (US\$ Mn)
Copper	0.00024	6.5	-0.001591	-7.31	0.000897	4.94
Nickel	0	0	0.000003	0	-0.000001	0
Tungsten	-0.00	0	-0.00	0	0.00	0

It may be seen that, except maybe in the case of copper, there possibly may not be a large impact on imports on account of a possible increase in imports of copper, nickel; and tungsten. Possibly in the case of copper, given the wide range of uses it has, there may be a positive impact on GDP on account of access to copper ores/concentrates at lower costs. However, even here, the possible impact may not be large, but it is still positive. In summary, a possible increase in imports these three minerals, majority demand of which is already being met through imports, may not have a negative impact on production, employment and the economy and possibly may have a probable positive impact, possibly on account of lower costs.

Simulation on Increase in Production and Select Imports

In the previous simulations, the possible impact of a probable increase in imports of those minerals, whose demand is majority met through imports was examined. However, there exists reserves for most of the minerals that are studied in this report and domestic production of the same also happens. To better understand the probable impact, assuming an increase in production happens, a simulation is undertaken wherein production of those minerals which are being produced domestically in industry is increased (in a simulation) to understand their possible impact on production, prices, employment, GDP and trade. Further, in this simulation, to also understand better, the possible cumulative impact imports of copper, tungsten and nickel may have; along with an increase in production of the minerals being produced in India, an increase in the imports of copper, tungsten and nickel is also considered. The results of this simulation are examined below. Further, unlike in the previous simulations, the supply of labor and capital are also increased in this simulation to better understand the possible impact of market dynamics.

Table 16: Change in Production due to increase in Production of 10 Minerals (incl coal) and Increase in Imports of Copper, Tungsten and Nickel (percentage change)

	Production	Prices	Exports	Imports
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>	<i>Column 4</i>	<i>Column 5</i>
Coal: mining and agglomeration of hard coal, lignite and peat	1.7	0.05	-0.09	-0.22
Oil: extraction of crude petroleum, service activities incidental to oil and gas extraction excluding surveying (part)	1.21	-0.01	0.95	0.68
Gas: extraction of natural gas, service activities incidental to oil and gas extraction excluding surveying (part)	1.26	0.04	-0.43	1.36
Iron ore	1.23	0.11	-0.03	0.2
Limestone	1.05	0.45	-0.57	0.12

Manganese	0.85	0.64	-0.9	0.19
Bauxite	1.04	0.46	-0.59	0.13
Lead	0.24	3.18	-4.74	1.9
Zinc	0.26	2.82	-4.2	1.59
Copper	-0.06	1	-1.43	1.47
Tungsten	0.12	118.86	-72.67	102.38
Gold	1.14	-0.01	0.14	0.29
Silver	1.24	0.2	-0.18	0.14
Rock Phosphate	0.13	-603104	-4.3E+08	-469778
Diamond	1.2	0.06	0.05	0.25
Nickel	0.12	144.05	-77.17	123.28
Manufacture of textiles	1.43	-0.58	4.06	-1.27
Manufacture of wearing apparel	2.16	-0.61	4.22	-1.7
Manufacture of leather and related products	2.35	-0.61	4.99	-1.21
Lumber: manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	0.82	-0.69	4.26	-1.64
Paper & Paper Products: includes printing and reproduction of recorded media	1.46	-0.55	2.86	-0.28
Petroleum & Coke: manufacture of coke and refined petroleum products	0.77	-0.06	0.53	0.46
Manufacture of chemicals and chemical products	1.68	-0.36	2.42	0.45
Manufacture of pharmaceuticals, medicinal chemical and botanical products	2.36	-0.63	3.94	0.03
Manufacture of rubber and plastics products	1.24	-0.51	3.27	-0.59
Manufacture of other non-metallic mineral products	0.21	-0.29	1.66	-0.81
Iron & Steel: basic production and casting	0.89	-0.42	2.21	-0.61
Non-Ferrous Metals: production and casting of copper, aluminium, zinc, lead, gold, and silver	0.97	-0.07	0.8	0.61
Manufacture of fabricated metal products, except machinery and equipment	0.68	-0.36	2.74	-0.99
Manufacture of computer, electronic and optical products	1.81	-0.39	3.58	-0.17

Manufacture of electrical equipment	1.39	-0.43	3.83	-1.01
Manufacture of machinery and equipment n.e.c.	1.07	-0.47	3.79	-1.25
Manufacture of motor vehicles, trailers and semi-trailers	0.72	-0.47	2.56	-1.01
Manufacture of other transport equipment	1.77	-0.48	3.99	-0.02
Other Manufacturing: includes furniture	1.03	-0.28	2.07	-0.75
Electricity; steam and air conditioning supply	0.95	-0.39	2.41	-0.23
Gas manufacture, distribution	1.06	-0.86	4.95	-1.24

Table 17: Change in Employment, Wages and Capital supply due to increase in Production of 10 Minerals (incl coal) and Increase in Imports of Copper, Tungsten and Nickel (percentage change)

	Factor Supply	Wages/Cost of Capital
<i>Column 1</i>	<i>Column 2</i>	<i>Column 3</i>
Clerks	0.73	-0.61
Service Shop/Factory Workers	0.71	-0.61
Technical/ Professional workers	0.98	-0.61
Officers/Managerial Professionals	0.74	-0.61
Agricultural and Other Low Skilled Workers	0.64	-0.61
Capital	1.18	-0.96

The underlying logic of this exercise being, either through policy measures or in response to lowering of production costs, there is an increase in supply of labor and capital. The lowering of production costs is being driven in this case by increase in an increase in production, employment and capital.

Looking at the impact on production (table 16, column 2), it is seen that, in response to a possible decrease in production costs, there may be an increase in the production of the entire manufacturing industry, including the labor-intensive industry (which previously did not exhibit much growth). A possible increase in the supply of the factors of production may drive this growth as it is seen that both labor and capital have increased their supply to meet the greater demand in the mining and manufacturing industry.

Turning to market prices (table 16, column 3) and wages (table 17), it is seen that in response to a possible increase in production across all industries, there may be a general a fall in prices in manufacturing, however there may be increase in price in mining. With respect to manufacturing, given that both the supply of labor and capital may have increased due to greater demand across all the industries, there may be a general decline in prices due to an expansion of the production base. But, there also may be a decline in the wages of labor and price of capital as there probably has been a general increase in the supply of labor and capital. But, the decrease in wages may be uniform across all the different categories, possibly showcasing a uniform decrease in wages, not heavily impacting one category of workers.

This decline in the prices of the industry may lead to greater exports (table 16, column 4). As may be seen here, the possible increase in the supply of labor, along lower production costs may help the manufacturing industry maintain price competitiveness and increase its exports. This particularly may be true for the labor-intensive industry such as textiles and apparel whose exports are driven, among other factors, by price competitiveness.

Turning now to the impact on real GDP, aggregate exports and imports (table 18), it is seen that real GDP has increased in this simulation. This indicates the positive feedback effect a possible increase in production and also possibly the increase in imports of minerals not produced in India may have on the economy. An expansion of production in mining, possibly leading to lower costs may be stimulating an increase in production. This alongside a possible increase in the supply of labor and capital maybe boosting export competitiveness, possibly leading to an increase in exports, which may also be reflected in the aggregate exports growth. Though imports are also increasing, this increase may have to do with a possible increase in consumption driven by decrease in costs and increase in exports (as many imports are used as intermediates in exports also). Further, the increase in imports is quite small.

Table 18: Change in real GDP, Aggregate Exports and Imports, due to due to increase in Production of 10 Minerals (incl coal) and Increase in Imports of Copper, Tungsten and Nickel (percentage change)

	Real GDP (% change)	% Change (US\$ Mn)	Aggregate Exports (% change)	% Change (US\$ Mn)	Aggregate Imports (% change)	% Change (US\$ Mn)
Impact	0.91	24106.75	2.40	11052.50	0.04	261.51

CHAPTER | 6

CONCLUSION

The findings of this study highlight the importance of adopting a balanced and forward-looking mineral policy. This chapter brings together key insights and suggests strategic approaches to strengthen self-reliance while ensuring continued global competitiveness.

WHAT'S IN THIS CHAPTER:

- Summary of key findings
 - Strategic policy recommendations
 - Insights for future planning
 - Scope for further research
-

This report attempts to understand what may be the possible impact of importing minerals on various aspects of the Indian economy. Thirteen minerals and coal were studied in this report.

It was seen that, India produces a wide range of minerals and metals, alongside having reserves for many of the minerals/metals. Further, given the similar geographical setup India has with many other mineral rich countries, the potential for continued expansion of the mining industry may exist.

With respect to import trends, it was seen that each of the mineral/metal identified in the study, both in terms of usage and import values, there was a lot of divergence between the identified minerals and also in the time series trend of each mineral itself. This constant fluctuation is also reflected in the CAGR growth rates of the minerals.

With respect to import duties, it was seen that, compared to the other top importers of minerals, at least in the case of the thirteen minerals studied in this report, India has the higher average customs duty in more or less all them. All of these minerals are further processed and used by the upstream and downstream industry for varied industrial/consumption purposes. It may be possible that, at the very least in minerals not being produced in India, the imposition of customs duties, may impact the downstream industry in terms of cost.

With respect to estimates of labor productivity as given in this report, it is seen that, as with import trends, there are wide fluctuations between the minerals, in terms of labor intensity of production, at least in the period 2011 – 2021. However, it may be mentioned that, at least from

2014, there is a consistent long term upward trend in the increase in labor productivity with respect to coal. A part of this reason may be attributed to reforms in coal auctioning, all of which had a positive impact on coal production, which is also being reflected in labor productivity with respect to coal.

With respect to the data on production, the data was sourced from the prowess database of CMIE and the total sales data of companies engaged in the mining activity of each of the minerals was used as a proxy for production data of each mineral. As this is company level data, it reflects the total sales of the company, engaged in the mining activities.

Turning now to the question of what may be possible impact of imports and production on the Indian mining industry, particularly for the identified minerals, in terms of mineral production, the manufacturing industry, employment, trade and the economy as a whole, a series of economic modelling simulations were undertaken. These simulations were computable general equilibrium in nature and were run on GTAP. Broadly speaking the simulations fall under two broad categories. Under the first category, in each different simulation, the imports of three minerals not being produced in India are conducted and its possible impact on production/supply of factors of production (labor and capital), prices/wages, tariff changes, exports, imports and real GDP was studied. In the second simulation, there was both an increase in production/factors of production and an increase in imports of those minerals not being produced in India.

With respect to the simulations where only imports of each different mineral was increased, it was found that the impact was quite minimal. A possible reason for the same may be that these minerals are already being imported into India and therefore an increase in their imports may not significantly impact production. Also, although minimal, there is a positive impact on GDP and employment.

Under the same category of simulations, it was seen that, there was decrease in the prices of minerals and those manufacturing industry which may use the minerals, possibly being more cheaply imported, more in their inputs than others, expanded production.

Again, under the same category of simulations (particularly for copper), it was also seen that, there was an increase in the wages of labor, due to increased production in some industries and there was also a fall in production of certain labor-intensive industries. This fall in production for labor intensive industries and general increase in wages, may be due to a greater demand for workers in those industries which are growing.

With respect to the simulation where production increased, supply of factors of production increased and there was an increase in imports of select minerals, it was seen that there was a general increase in the supply of labour and capital.

It was also seen that, all the different manufacturing industries experienced growth, on account of greater access to capital and labor. It was further seen that there was a general decline in prices across the manufacturing industry, on account of increased production.

It was also seen a possible increase in tariffs may also drive increase in production of those minerals having production/reserves in India. This increase in tariffs may possibly explain the rise in prices for those minerals.

Unlike, in the simulations where only imports may increase, in the one where production may increase, exports may also increased, both at an industry level and at an aggregate level, possibly on account of increased production and decreased prices. Although imports have increased across all the simulations, this increased may be on account of higher demand, greater consumption and due to increase in exports, greater demand for intermediates in exported products. Further, as seen, the rise in imports is quite small as compared to exports, indicating the possibly benefit, increased mineral and manufacturing production may have on trade.

Overall, across all the scenarios, it was seen that there was a positive change in real GDP. The scenario where both production and supply of factors of production increased recorded the highest percentage growth in real GDP.

The impact of imports, although uneven, has for the most part have a positive impact on GDP and employment with the primary channel of impact being lower costs of production.

In summary, it may be mentioned that, continued efforts (in terms of policy changes, encouraging more mineral production) to increase the productive capacity of the mining and manufacturing industry may have a greater impact on generating employment, exports and increasing economic growth.

Appendix I: The Global Trade Analysis Model (an overview)

The Global Trade Analysis Project (GTAP) Data Base provides a comprehensive overview of the global economy through a detailed representation of bilateral trade international transportation margins and domestic economic flows. This recently updated database serves as the foundation for computable general equilibrium modelling and details the annual value of economic activities across 145 individual countries and 18 composite regions. The model comprises of 65 discrete goods and services sectors evaluated at pre-tax and post-tax valuations over seven reference years from 2004 to 2023 (Aguilar et al 2026), with a corresponding GTAP model (Hertel, 1997).

Essentially this model was designed with the objective of making it easier for researchers who aim to derive quantitative estimates of several economic parameters on a country/regional level. Its data sources act as a representation of the global economy and are thus used as the initial equilibrium for the standard modelling framework.

The construction of this expansive database relies heavily on the accretion and sourcing of input output tables for each country that is also supplemented by internationally sourced macroeconomic data. The primary macroeconomic condition enforced throughout the data reconciliation procedure guarantees that gross domestic product equals the sum of private consumption investment government consumption and net exports. Researchers utilize data from the World Development Indicators to constrain national accounts in millions of current United States dollars and impose consistent market exchange rates across all reference years.

Essentially, GTAP assumes parity between savings and investment so that the trade balance is equal to savings minus investment net of depreciation in order to maintain accounting identities across diverse datasets. Another assumption is pegging export and import targets and then coercing bilateral trade data in the database as way of adjusting domestic expenditure aggregates. Thus, trade data is used to adjust or simulate changes in Gross Domestic Product (GDP) while leaving the level of savings to be computed as a residual.

International trade data reconciliation forms another foundational pillar of the database architecture. The estimation of merchandise trade relies on United Nations commodity trade statistics where an advanced methodology reconciles frequent discrepancies between reported imports and exports to guarantee balanced global trade.

This procedure ensures that global exports strictly align with global imports for each commodity while domestic exports are carefully derived by converting total imports into retained imports to accurately reassign re-exports to their true country of origin. The treatment of services trade has similarly evolved to incorporate balanced bilateral data sourced from the Organisation for Economic Co-operation and Development (OECD) and the World Trade Organization (WTO).

This dataset provides time series statistics classified into numerous service categories based on extended balance of payments definitions. In cases where these broad categories do not map directly to the specific service sectors of the database, the researchers apply disaggregation procedures utilizing trade in services data delineated by mode of supply. This careful matching process creates the baseline of domestic sales global trade and shipping costs needed to build accurate economic models.

Border protection and domestic tax policies undergo extensive processing to accurately reflect the intricate global policy environment. The database supersedes basic tax information contained within contributed input output (IO) tables by integrating comprehensive border protection data compiled from specialized international organizations. Bilateral import tariffs are meticulously incorporated for all merchandise products utilizing applied ad valorem tariffs and the equivalents of specific tariffs and import quotas provided by the International Trade Centre. For agricultural sectors the database captures domestic support measures using producer support estimates from the OECD alongside export subsidy notifications collected from the WTO.

These sophisticated tariff matrices exclude temporary trade remedies such as anti-dumping measures countervailing duties or safeguard mechanisms. Researchers intending to model these specific policy instruments must subsequently adjust the baseline tariffs within their computable general equilibrium framework. Through this detailed accounting of protectionist instruments the model allows economists to rigorously simulate complex policy scenarios including unilateral tariff impositions bilateral escalations and comprehensive multilateral tariff eliminations.

Energy data and environmental emissions accounts are heavily integrated to support contemporary climate and resource policy analyses. Energy volumes are primarily represented using extended energy balances developed by the International Energy Agency complemented by United Nations energy statistics to maximize geographic coverage (IEA 2023). Bilateral energy trade flows are further refined using specialized statistical reviews to address known weaknesses in standard commodity trade datasets. To convert these physical energy volumes into precise value flows the database relies on domestic energy prices and taxes complemented by export and import prices of fossil fuels. Substantial modifications have been introduced to the accounting of carbon dioxide emissions from fossil fuel combustion by implementing the more rigorous methods outlined in the 2006 guidelines of the Intergovernmental Panel on Climate Change (IPCC, 2006). This framework includes the precise estimation of emission factors at a highly granular commodity level alongside a transparent treatment of emissions from industrial processes like flaring and blast furnaces. The database is additionally supplemented by complementary accounts detailing non carbon dioxide greenhouse gases and air pollutant emissions mapped to specific industrial output endowment and input uses.

Scholars exploring global economic networks frequently utilize similar multi regional input output databases to evaluate international trade and environmental footprints. The World Input Output Database provides a comparable structural framework consisting of a series of global tables

covering major world economies to analyze the anatomy of the global trade slowdown and trace global value chains. This resource maintains detailed socio economic and environmental accounts enabling robust structural decomposition analyses of energy transitions and carbon emissions embodied in international commerce. Researchers also rely extensively on EXIOBASE which constructs a detailed time series of environmentally extended multi regional input output tables designed explicitly for environmental economic accounting and consumption based footprint assessments. This database utilizes rectangular supply and use tables as primary building blocks to achieve an exceptionally high sectorial detail matched with multiple social and environmental satellite accounts. Finally the Eora global multi region input output database offers an alternative framework optimized for high country and sector resolution mapping the structure of the world economy across thousands of sectors. This highly detailed database contains extensive environmental indicators covering greenhouse gas emissions water requirements and land occupation to facilitate comprehensive analyses of international supply chains and resource efficiency (Lenzen et al 2013; Tukker et al 2015; Timmer et al 2017).

Appendix II: Description of Products

A total of thirteen minerals and coal were identified for the study. These minerals were mapped to their HS codes (HS 6 digit) and were further categorized into their end-use categorization, processing stage and broad industry categorization. The same are summarized in the table below

The first point that stands out is that almost all the products are categorized as intermediate goods, indicating that all of them are used further in the value chain, as is befitting of minerals. For the goods that are categorized with dual use, there are categorized under intermediate/final consumption.

With respect to their processing stage, a majority of the products are classified as primary meaning that they are at the beginning of processing/value chain. The few that are classified under the processed category have undergone some form of change from their initial extraction as ores. The minerals classified as processed include gold, diamond, limestone, and silver.

In terms of industry categorization, almost all the identified minerals are classified under broad categories of mining and quarrying (separated only in description by the type of metal being mined). The only minerals/metals which are categorized separately are those which may be termed as ‘precious metals/minerals’, i.e. gold, silver and diamond. This separate categorization indicates that at least for these three metals/minerals, processed/semi processed imports happen, apart from any imports of ores.

Table 19: Mapping of Minerals with HS Code, End Use and Industry categorization

Mineral	HS Code	Product Description	End Use Categorization	Processing Stage	Industry Categorization
Iron Ore	260111	Non-agglomerated iron ores and concentrates (excluding roasted iron pyrites)	Intermediate	Primary	Mining of iron ores
	260112	Agglomerated iron ores and concentrates (excluding roasted iron pyrites)	Intermediate	Primary	Mining of iron ores

Limestone	252100	Limestone flux; limestone and other calcareous stone	Intermediate	Processed	Quarrying of stone, sand and clay
Manganese	260200	Manganese Ore (46% and Above)	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Bauxite	260600	Bauxite (Natural) not Calcined	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Lead	260700	Lead Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Zinc	260800	Zinc Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Copper	260300	Copper Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Tungsten	261100	Tungsten Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Gold	261690	Gold Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores

	710811	Gold, incl. gold plated with platinum, for non-monetary purposes	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals
	710812	Gold, incl. gold plated with platinum, unwrought, for non-monetary purposes (excluding gold ...	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals
	710813	Gold, incl. gold plated with platinum, in semi-manufactured forms, for non-monetary purposes	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals
Silver	261610	Silver Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
	710610	Powder of silver, incl. silver plated with gold or platinum	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals
	710691	Silver, incl. silver plated with gold or platinum, unwrought (excluding silver in powder form)	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals
	710692	Silver, incl. silver plated with gold or platinum,	Intermediate	Processed	Manufacture of basic precious and non-ferrous metals

		semi-manufactured			
Rock Phosphate	251010	Natural Calcium Phosphates Un-ground	Intermediate	Primary	Mining of chemical and fertilizer minerals
	251020	Natural Calcium Phosphates Ground	Intermediate	Primary	Mining of chemical and fertilizer minerals
Diamond	710210	Unsorted diamonds	Intermediate/Final Consumption	Processed	Other mining and quarrying n.e.c.
	710221	Industrial diamonds unworked or simply sawn, cleaved or bruted	Intermediate	Processed	Other mining and quarrying n.e.c.
	710231	Unworked or simply sawn, cleaved or bruted	Intermediate/Final Consumption	Processed	Other mining and quarrying n.e.c.
	710239	Diamond, cut or otherwise worked but not mounted or set	Intermediate/Final Consumption	Processed	Manufacture of jewellery and related articles
Nickel	260400	Nickel Ores and Concentrates	Intermediate	Primary	Mining of non-ferrous metal ores, except uranium and thorium ores
Coal	270111	Anthracite Coal whether or not Pulverised but not Agglomerated	Intermediate	Primary	Mining and agglomeration of hard coal
	270112	Bituminous Coal whether or not Pulverised but not Agglomerated	Intermediate	Primary	Mining and agglomeration of hard coal

	270119	Coking Coal	Intermediate/Final Consumption	Primary	Mining and agglomeration of hard coal
	270119	Steam Coal	Intermediate/Final Consumption	Primary	Mining and agglomeration of hard coal
	270400	Coke and Semi-Coke of Lignite or of Peat	Intermediate/Final Consumption	Primary	Manufacture of coke oven products

Source: ITC Trade Map and Broad Economic Classification

Appendix III: Data Source

As the study concerns itself with the imports of minerals, the data thus collected primarily focused on imports of ores/concentrates/fines. Tariff subheadings for the minerals listed below were identified:

15. Coal
16. Iron Ore
17. Limestone
18. Manganese
19. Bauxite
20. Lead
21. Zinc
22. Copper
23. Tungsten
24. Gold
25. Silver
26. Rock Phosphate
27. Diamond
28. Nickel

The tariff subheadings for the most part reflected imports of the minerals in ores/concentrates form, except for gold, diamond and silver, for which three, tariff subheadings under which imports happen at the semi-processed form. This is because a majority of imports of the two precious metals and diamond happen under those tariff sub-headings.

Since the study also concerns itself with the impact on the economy, production data and employment data, or rather proxies of it were also captured. Data as extracted from the prowess data of Centre for Monitoring the Indian Economy (CMIE) was used for this exercise. Data on total sales, compensation to employees and number of employees was used as proxies for production, labor costs and labor productivity.

The CMIE data was mapped to relevant National Industrial Classification (NIC) and the NIC codes were mapped to the tariff sub-headings under the Harmonized Standard Nomenclature (HSN) to ensure there is minimal discrepancy between the trade and domestic data.

It may be mentioned that for certain metals (gold, silver, tungsten, nickel, zinc etc.) NIC does not give separate codes but have clubbed two metals under a single category (for example: gold & silver come under a single code). It may also be mentioned that no comparable sales data was

found for rock phosphate in prowess data. Further, with respect to diamond, the closest data found in prowess was with respect to manufacturing/processing of diamond. Since diamond is imported in a semi-processed format, this may be adequate reflection.

Lastly, the data reported is for companies engaged in the mining industry. While data for only those companies engaged in mining of the concerned minerals in question is taken, the sales, employment and compensation data is reflective of the company as a whole and not only for that part of the company engaged in the mining/processing of the minerals that are studied in this report. The data, as such is taken, has no separate accounts/data available for production/sales of the concerned companies for only the minerals in question.

The full list of tariff sub-headings, NIC codes and CMIE product industry code are given below.

Table 20: List of Tariff sub-heading, NIC Codes and CMIE Product industry code

Mineral	HS Code	HS Code Product Description	NIC Code	NIC Code Product Description	CMIE Product Industry Code
Iron Ore	260111	Non-agglomerated iron ores and concentrates (excluding roasted iron pyrites)	7100	Mining of iron ores	27080400000000000000
	260112	Agglomerated iron ores and concentrates (excluding roasted iron pyrites)			
Limestone	252100	Limestone flux; limestone and other calcareous stone	8107	Mining/quarrying of limestone, limeshell, 'kankar' and other calcareous minerals including calcite, chalk and shale	27046300000000000000

Manganese	260200	Manganese Ore (46% and Above)	7293	Mining of manganese ore	27081200000000000000
Bauxite	260600	Bauxite (Natural) not Calcined	7292	Mining of aluminum ore (bauxite)	27082800000000000000
Lead	260700	Lead Ores and Concentrates	7296	Mining of lead and zinc ore	27088004000000000000
Zinc	260800	Zinc Ores and Concentrates	7296	Mining of lead and zinc ore	27088004000000000000
Copper	260300	Copper Ores and Concentrates	7291	Mining of copper ore	27087600000000000000
Tungsten	261100	Tungsten Ores and Concentrates	7299	Mining of other non-ferrous metal ores, n.e.c. [titanium (ilmenite and rutile) ,niobium, tantalum, vanadium or zirconium ores; tin bearing ores; and, nickel, cobalt, tungsten, molybdenum, antimony and other non-ferrous metal ores]	27084000000000000000
Gold	261690	Gold Ores and Concentrates	7295	Mining of precious metal ore (gold, silver)	27086000000000000000
	710811	Gold, incl. gold plated with platinum, for non-monetary purposes			
	710812	Gold, incl. gold plated			

		with platinum, unwrought, for non-monetary purposes (excluding gold ...			
	710813	Gold, incl. gold plated with platinum, in semi-manufactured forms, for non-monetary purposes			
Silver	261610	Silver Ores and Concentrates	7295	Mining of precious metal ore (gold, silver)	27086000000000000000
	710610	Powder of silver, incl. silver plated with gold or platinum			
	710691	Silver, incl. silver plated with gold or platinum, unwrought (excluding silver in powder form)			
	710692	Silver, incl. silver plated with gold or platinum, semi-manufactured			
Rock Phosphate	251010	Natural Calcium			

		Phosphates Un-ground			
	251020	Natural Calcium Phosphates Ground			
Diamond	710210	Unsorted diamonds	32112	Working of diamonds and other precious and semi- precious stones including the working of industrial quality stones and synthetic or reconstructed precious or semi- precious stones	54240612000000000000
	710221	Industrial diamonds unworked or simply sawn, cleaved or bruted			
	710231	Unworked or simply sawn, cleaved or bruted			
	710239	Diamond, cut or otherwise worked but not mounted or set			
Nickel	260400	Nickel Ores and Concentrates	7299	Mining of other non-ferrous metal ores, n.e.c. [titanium (ilmenite and rutile) ,niobium,	27084000000000000000

				tantalum, vanadium or zirconium ores; tin bearing ores; and, nickel, cobalt, tungsten, molybdenum, antimony and other non-ferrous metal ores]	
Coal	270111	Anthracite Coal whether or not Pulverised but not Agglomerated	0510	Mining of hard coal	27120404000000000000
	270112	Bituminous Coal whether or not Pulverised but not Agglomerated			
	270119	Coking Coal			
	270119	Steam Coal			
	270400	Coke and Semi-Coke of Lignite or of Peat			

Source: Authors calculations based on ITC Trade Map, NIC (2008) and Prowess data

The data as taken from prowess has missing data points. This is applicable across all the minerals studied in the report. The missing data points are reflective of company data that is missing for those years. Data with respect to employment has the greatest number of missing variables. For a few years, data on sales, compensation and number of persons employed is also missing. All of this, may have a bearing on the results.

A summary of the total number of data points, across all companies, with respect to the minerals studied in this report, for the combined period 2011 to 2021, with respect to sales, compensation to employees and number of employees is given below.

Table 21: Aggregate of Data Points (Total, Available & Missing), 2011 to 2021

Mineral/Metal	Variable	Total Data	Available Data	Missing Data	Percentage Missing
Iron Ore	Sales	330	227	103	31.21
	Compensation to Employees	330	253	77	23.33
	No of Employees	330	94	236	71.52
Diamond	Sales	1386	849	537	38.74
	Compensation to Employees	1386	888	498	35.93
	No of Employees	1386	204	1182	85.28
Manganese	Sales	341	205	136	39.88
	Compensation to Employees	341	230	111	32.55
	No of Employees	341	48	293	85.92
Gold & Silver	Sales	561	319	242	43.14
	Compensation to Employees	561	331	230	41.00
	No of Employees	561	101	460	82.00
Bauxite	Sales	231	154	77	33.33
	Compensation to Employees	231	152	79	34.20
	No of Employees	231	52	179	77.49
Limestone	Sales	517	306	211	40.81
	Compensation to Employees	517	301	216	41.78
	No of Employees	517	81	436	84.33
Coal	Sales	2486	1702	784	31.54
	Compensation to Employees	2486	1784	702	28.24
	No of Employees	2486	381	2105	84.67
Lead & Zinc	Sales	22	19	3	13.64

	Compensation to Employees	22	19	3	13.64
	No of Employees	22	6	16	72.73
Copper	Sales	11	10	1	9.09
	Compensation to Employees	11	10	1	9.09
	No of Employees	11	7	4	36.36
Tungsten & Nickel	Sales	11	7	4	36.36
	Compensation to Employees	11	6	5	45.45
	No of Employees	11	0	11	100.00

Source: Authors calculations based on Prowess data

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ABOUT THE BOOK

This study presents a comprehensive and data-driven examination of India's mineral import landscape, offering a distinctive economy-wide perspective rarely found in existing literature. By integrating long-term trade trends with advanced simulation and modelling techniques, it moves beyond conventional descriptive analysis to evaluate the real economic implications of mineral import decisions on output, employment, prices, and trade dynamics. Covering a wide spectrum of critical minerals and situating India within the global resource ecosystem, the study provides a balanced and policy-relevant framework for understanding the interplay between domestic production and strategic imports. Its analytical depth, empirical rigor, and practical relevance make it a valuable resource for policymakers, researchers, and industry stakeholders engaged in shaping India's resource and industrial strategy.

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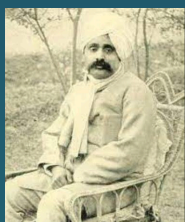
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Born in the Palnadu region of present-day Andhra Pradesh, he was a courageous farmer leader. He led a strong resistance against oppressive British taxation during the colonial era. Standing with rural communities, he refused to submit to unjust authority. His ultimate sacrifice became a symbol of farmers' dignity and fearless defiance. He remains an enduring inspiration of courage and justice in India's freedom struggle.



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